Filling the Void: How Competing Brands Can Capitalize on a Brand Deletion

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ABSTRACT

Brand manufacturers, big and small, regularly prune their brand portfolios by deleting their brands from categories. Competing brands try to fill the void, with potentially varying consequences for sister brands by the same manufacturer, rival national brands, and private labels. As little is known about these consequences, this research analyzes systematically whether these three brand types benefit and how they adjust their marketing conduct following a brand deletion. Using a difference-in-differences analysis of 1,046 FMCG national brand deletions observed over 10 years affecting competing brands sold across 201 US markets and 55 retailers (with about 63,000 market-retailer-category-brand combinations affected), the findings reveal that the outcomes of the deletions are strongly shaped by how brands adjust their marketing mix after the brand deletion. By lifting distribution (which is highly effective), private labels are the largest beneficiaries of brand deletions, followed by sister brands (successfully lifting line length), and, lastly, rival national brands. There are also lost opportunities (little change in price promotion across brand types, spoiled arms (hike in ineffective feature advertising), and self-inflicted losses (regular price hikes that decrease revenue). Manufacturers retain only 38% of the deleted brand's revenue through sister brands on average, showing the risk of brand pruning.

Keywords: Marketing Mix, Brand Management, Marketing Strategy

Brand manufacturers, big and small, regularly engage in pruning by deleting brands from categories. In their efforts to manage their extensive brand portfolios across many product categories, often with multiple brands within the same category, P&G announced in 2014 plans to shrink its brand portfolio by more than half by retiring as many as 100 brands (Wall Street Journal 2014). Coca-Cola indicated it will discontinue 200 brands, including its Tab, Zico, and Odwalla beverage brands (Wall Street Journal 2020). Not just global conglomerates, but also smaller brand manufacturers frequently delete brands from their portfolios. Brand manufacturers mention a variety of reasons for brand deletions, ranging from brands underperforming or having low growth potential (Crespo 2018), brand portfolio simplification and increased efficiencies (Morning Brew 2020), reducing overlap in targeted customer segments (Rose 2025), to changing consumer preferences resulting in shrinking categories (Unilever 2025).

Such deletions offer several strategic benefits, including cost savings (e.g., in procurement, production, inventory, distribution, salespeople, and opportunity costs), greater efficiency, and a sharper focus on core brands. In addition, brand manufacturers hope that some customers of the deleted brand will simply switch to "sister brands" within the same manufacturer's portfolio (Kumar 2003), if such sister brands exist. However, the extent to which these sister brands (if available) actually benefit from deletions has not been documented though, or what the effects might be for other brands in the market, such as rival national brands (owned by other manufacturers) and, if available, private labels (owned by retailers). It is also not clear what manufacturers of the different brands can do to reap these benefits.

Arguably, one brand's demise could benefit other brands in the product category, which may capture the deleted brand's business. Thus, we study changes in the marketing conduct of sister brands, rival national brands, and private labels following a focal brand's exit from a

product category. In the context of FMCG (our research context), marketing conduct includes manufacturer-driven decisions (feature advertising and price promotion), retailer decisions with manufacturers' input (regular price and line length, i.e., the number of SKUs that a brand carries), and retailer-driven decisions (distribution).[1] Competing brands may strengthen their marketing support to seize the opportunity offered by the deleted brand's exit, or weaken it because competition has been softened. Both changes can be considered as responses to the brand deletion, as long as they involve statistically significant changes in competing brand marketing conduct when a brand is deleted compared to the case where it is not deleted.

While extant marketing literature offers plenty of guidance for new brand launches, there is very little consideration of how competing brands might fill voids left behind by another brand's permanent deletion from all retailers. Although the literature has focused on temporary SKU removals (e.g., Campo, Gijsbrechts, and Nisol 2000; Campo, Gijsbrechts, and Nisol 2003; Campo, Gijsbrechts, and Nisol 2004; Sloot, Fok, and Verhoef 2006) and permanent SKU deletions (e.g., Boatwright and Nunes 2001; Broniarczyk, Hoyer, and McAlister 1998; Sloot et al. 2006), research has not studied deletions where a brand is deleted at all retailers. Instead, other papers have looked at the implications of temporary out-of-stock situations (Verbeke, Farris, and Thurik 1998), conflict delistings (Van der Maelen, Breugelmans, and Cleeren 2017), and the deletion of a brand at one but not at all retailers (Sloot and Verhoef 2008).

To the best of our knowledge though, no research has dealt with permanent brand deletions from a product category across all retailers, which constitute a deliberate strategic decision to withdraw the brand permanently from the product category and from all retailers in a market. As such, permanent brand deletions differ fundamentally from (1) out-of-stock situations, which are

[1] In other contexts (e.g., digital marketing), marketing conduct includes other marketing activities, e.g., through social media, banner ads, or email.

not deliberate and not permanent; (2) conflict deletions, which do not apply to the entire market and are not permanent, (3) SKU deletions, which are tactical decisions, (4) adoption of an exclusive distribution strategy, where the brand is only sold at select retailers, and (5) the deletion of a brand at one but not at all retailers.

In making strategic brand deletion decisions, all SKUs of a brand disappear from a category across all retailers, making these decisions highly noticeable and relevant to consumers and competitors. To understand the performance implications of brand deletions, we need empirical insights into which remaining brands benefit. If sister brands benefit exclusively, it supports brand pruning for the manufacturer, as revenues shift within the portfolio rather than being lost. However, if rival national brands benefit and little goes to sister brands, the manufacturer of the deleted brand may regret its brand-pruning strategy. Finally, if private labels benefit, the brand deletion shifts more power to retailers. As these hypothetical outcomes imply, assessing which brand benefits the most from brand deletions has crucial ramifications for the manufacturer, its rivals, and its retailer partners.

We also need to consider if and how brands may leverage their marketing mix to ensure that redistributed demand comes to them. For example, a brand could offer price promotions, use feature advertising, expand its product line, increase distribution, and/or adjust its regular price. Rival national brands may undertake such marketing efforts in the wake of the brand deletion, but so could sister brands from the same manufacturer, aiming to keep demand "in the family." Retailers may also see an opportunity to capitalize on the deleted brand's demise by pushing their own private labels. Thus far, "little is known about how marketing mix actions … can be used to mitigate the negative consequences of product unavailability" (Breugelmans, Gijsbrechts, and Campo 2018, p. 217). Understanding how such marketing efforts affect redistribution

outcomes is crucial for designing an informed response to a brand deletion from a category.

To document what happens in the wake of a brand deletion, this paper addresses four pertinent research questions. First, when a brand leaves the category, which competing brand type gains the most in terms of revenue: sister national brands, rival national brands, or private labels, and how is category revenue affected? Second, how are these gains moderated by the characteristics of the deleted brand (e.g., pre-deletion market share), the competing brand (e.g., similarity in price positioning to deleted brand), or the category (e.g., concentration)? Third, to what extent do competing brands change their marketing conduct significantly in the wake of the brand deletion? Fourth, how do competing brands fare overall after the brand deletion, taking into account the extent to which they engage in marketing mix adjustments?

To address these questions, we adopt an empirics-first (EF) approach (Golder et al. 2023) rather than a theory-first approach because little theory predicts post-deletion outcomes. We document empirical regularities, using U.S. scanner data over 10 years with 1,046 national brand deletions, from 201 designated market areas (DMAs), and within the top 3 retailers in each market. The deletions affect 6,031 brands competing in the same categories, resulting in over 63,000 market-retailer-category-brand combinations.

We use a stacked difference-in-differences approach with "not-yet-treated" cases as control units, combined with instrumental variables to account for potential endogeneity in the brand deletion timing and marketing mix. Using this setup, we find that a brand deletion benefits competing brands, increasing their sales by 1.73% on average. But not all brands benefit equally: retailers' private labels benefit the most (+3.45% revenue), followed by sister brands (+2.77%) and rival national brands (+1.36%). Competing brands gain more revenue when they have greater price similarity with the deleted brand, when the deleted brand's pre-deletion market

share is higher, and when the category is less concentrated. A large part of these revenue gains is attributable to marketing mix adjustments. All brand types manage to increase their distribution in the wake of the brand deletion, which, when combined with strong distribution elasticities, leads to revenue gains of 1.23%-1.77%. One surprising result is that only sister brands increase their line length in the wake of the brand deletion, which (combined with a strong line length elasticity) generates an additional .84% revenue gain. We also find evidence for *lost opportunities* in marketing conduct (hardly any change in the use of very effective price promotion), *self-inflicted losses* (regular price hikes even though this hurts brand revenue), and *spoiled arms* (lift in relatively ineffective feature advertising compared to the control group). We also find that on average, manufacturers retain only about 38% of the deleted brand's sales, with strong variation in this percentage across cases.

RELATED LITERATURE

A stream of prior literature focuses on what happens when a product, SKU, or brand is temporarily delisted or permanently deleted. As Table 1 shows, most of these studies focus on cases where the brand remains available but specific SKUs are temporarily delisted or permanently deleted.

Temporary brand delistings have been studied in the context of manufacturer–retailer conflicts and stockouts. For example, Van der Maelen et al. (2017) demonstrate that both manufacturers and retailers experience sales losses during conflict delistings. Although retailers are hit harder during the delisting period, they often recover, while manufacturers emerge stronger and enjoy higher post-delisting sales. In contrast, Hermans, Raassens, and Cleeren (2024) document severe negative effects of conflict delistings on the stock market value of both manufacturers and retailers. They also show how the effectiveness of advertising and price

promotions varies with conflict characteristics such as elimination size, publicity, and initiator. Vafainia et al. (2025) take a broader view of conflict delistings by studying not only involved brands and categories, but also uninvolved brands, categories, and even uninvolved retailers, and how price and promotion adjustments can shape these temporary cross-category spillover effects.

Table 1. Selected Studies of SKU and Brand Delistings (Temporary) and Deletions (Permanent)

Temporarily unavailable	**Product/SKU Delisting**	**Brand Delisting**
	Campo et al. (2000)	**Conflict delisting**
	Campo et al. (2003)	Hermans et al. (2024)
	Campo et al. (2004)	Van der Maelen et al. (2017)
	Emmelhainz, Stock, and Emmelhainz (1991)	Vafainia et al. (2025)
	Fitzsimons (2000)	**Out-of-stock**
	Kalyanam, Borle, and Boatwright (2007)	Peckham (1963)
	Peckham (1963)	Verbeke et al. (1998)
	Schary and Christopher (1979)	
	Sloot, Verhoef, and Franses (2005)	
	Verhoef and Sloot (2009)	
	Walter and Grabner (1975)	
	Zinn and Liu (2001)	
Permanently deleted	**Product/SKU Deletion**	**Brand Deletion at One Retailer**
	Argouslidis and Baltas (2007)	Sloot and Verhoef (2008)
	Boatwright and Nunes (2001)	
	Borle et al. (2005)	
	Broniarczyk et al. (1998)	
	Campo et al. (2004)	
	De Clerck et al. (2001)	**Brand Deletion at All Retailers**
	Drèze, Hoch, and Purk (1994)	
	Dukes, Geylani, and Srinivasan (2009)	This paper
	Gourville and Soman (2005)	
	Harness and Marr (2001)	
	Homburg, Fürst, and Prigge (2010)	
	Oppewal and Koelemeijer (2005)	
	Sivakumar and Raj (1997)	
	Sloot et al. (2006)	
	Tan and Cadeaux (2011)	
	Zhang and Krishna (2007)	

Rather than temporary brand deletions that are typically restricted to just one retailer, we study permanent brand deletions across all retailers. Since our focal brand deletions occur across all retailers, the deletion is more of a manufacturer's decision than a decision by a specific

retailer. Retailers and manufacturers tend to be rather silent on permanent brand deletions, which is different from conflict delistings that are widely discussed in the (business) press and can be observed in-store by customers through the empty shelves with notifications about the delistings.

Our study also differs from Sloot and Verhoef (2008, SV2008 from now on) studying the case where a brand disappears permanently at some but not all retailers, which allows customers to find the brand elsewhere. SV2008 uses an online lab experiment and a survey to study the impact of brand deletions on store and brand switching *intentions*, moderated by the deleted brand's characteristics and category characteristics. Our study extends SV2008 by studying brand deletions across all retailers (not just at one retailer) and by going beyond intentions and measuring the revenue effects of brand deletions for competing brands and the category.

Another point of difference is that SV2008 does not measure which competing brands benefit from the brand deletion, nor how these brands change their marketing conduct. What our paper adds to the literature is that we study which types of competing brands (rival national brands, sister brands, private labels) gain from the brand deletion, what moderates these gains, how marketing conduct changes in the wake of the brand deletion, and how competing brands fare overall after the brand deletion, taking into account their marketing conduct.

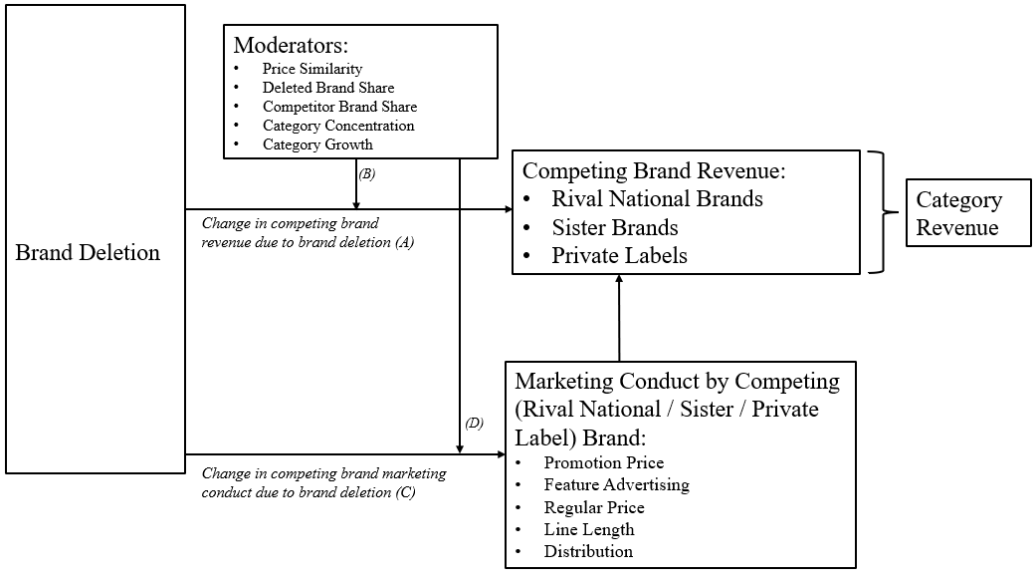
STUDY FRAMEWORK

We present the study framework in Figure 1. We purposefully refrain from directional hypotheses but include the figure to help visualize the focal relationships, labelled (A)-(D). Specifically, Figure 1 depicts the brand deletion as the focal event, with potential revenue gains for rival national brands, sister brands, and private labels (A). The literature on brand switching offers insights into which type of brands may benefit most. Since national brands compete more with other national brands rather than private labels (e.g., Blattberg and Wisniewski 1989), we

may expect rival national brands and sister brands to benefit the most. We also examine how brand deletion affects category revenue, which may be negative (if the deleted brand's demand is not fully absorbed by competing brands), neutral (if others gain as much as is lost), or positive (if competing brands expand the category).

Next, we study *moderators* shaping these gains (B in Figure 1). In particular, the larger the deleted brand (as indicated by its market share), the greater the potential gains for competitors (Sloot and Verhoef 2008). The competing brand's market share may also influence gains, positively if larger brands have more pull, or negatively due to ceiling effects (Fok and Franses 2004). Competing brands may be better positioned to grow revenue if their price positioning aligns more closely with that of the deleted brand (Blattberg and Wisniewski 1989). The other two moderators are category growth (which should lift the gain as the pie is growing) and category concentration (which may reduce the gain as it may be harder to find a suitable substitute in a category with a few big brands).

Figure 1. Study Framework



Notes: Sister brands refer to brands owned by the same manufacturer as that of the deleted brand, rival national brands refer to brands owned by other manufacturers, and private labels refer to retailer-owned brands.

Each competing brand can adjust its marketing conduct (captured by promotion price, feature advertising, regular price, line length, and distribution in our FMCG context[2]) to steer more revenue its way after a brand deletion (C in Figure 1). The direction of change can go either way. When the focal brand is deleted, competition softens, and competing brands may reduce marketing support (fewer price promotions or features, higher regular price, shorter line length, and less distribution). Conversely, the deleted brand's exit leaves a gap that competitors may seek to fill by increasing their marketing support. Finally, we explore how the same moderators shape these changes in competing brands' marketing conduct after a brand deletion (D in Figure 1). For instance, depending on the deleted brand's market share or the category concentration, competing brands may respond more aggressively in adjusting their marketing conduct.

DATA AND MEASURES

Deleted Brands

For a broad investigation of what happens in the wake of a brand deletion, we focus on all national brand deletions across all FMCG categories across a ten-year time period (2010–2019) from any brand manufacturer, ranging from small to large. We hence do not consider private label deletions. We identify brand deletions based on RMS NielsenIQ scanner data at the market-retailer-category-brand-week level. These scanner data cover more than 1,300 product categories in nine departments, such as health and beauty aids, food and beverages, dairy, and non-food grocery products. We adopt the NielsenIQ definitions of categories, and give a complete list of categories in our sample in Web Appendix A. We assume that all brands within a category can be alternatives to the deleted brand, and if they are not, this will show up empirically as null effects. As our focus is on brand deletions (rather than sub-brand deletions,

[2] Brands often carry a narrower range of products at a specific retailer in a specific market than across all retailers and markets in the country. Thus, it is relatively easy for them to extend their line length at a specific retailer on short notice.

which is more similar to previously-studied product line deletions), we aggregate NielsenIQ's sub-brands (e.g., "Air Wick Double Fresh", "Air Wick Essential Mist", "Air Wick Fresh Matic", and "Air Wick Hidden Pleasures") to the brand level ("Air Wick"). We decide to keep the NielsenIQ definitions of categories because they cater to different segments and have their own price levels (and at times different measurement units) and differentiated marketing activities.

We screen all product categories and identify deletions from any category for any national brand. We define a brand as deleted if it disappears from a product category and is eventually deleted across all markets (DMAs) in the U.S and does not reappear for at least one year. To ensure sufficient economic substance, we focus on deleted brands with (i) at least 52 weeks of observations, (ii) at least $10k in revenue across the U.S. in the year prior to the deletion, and (iii) at least $100 in weekly sales at a given retailer and market before the deletion. Thus, we identify 1,046 deleted national brands and show a selection in Table 2. Figure 2 provides an overview of the distribution of the deleted brands' market shares prior to the beginning of the deletion.

Figure 2. Histogram of Deleted Brands' Market Shares (N = 14,462)

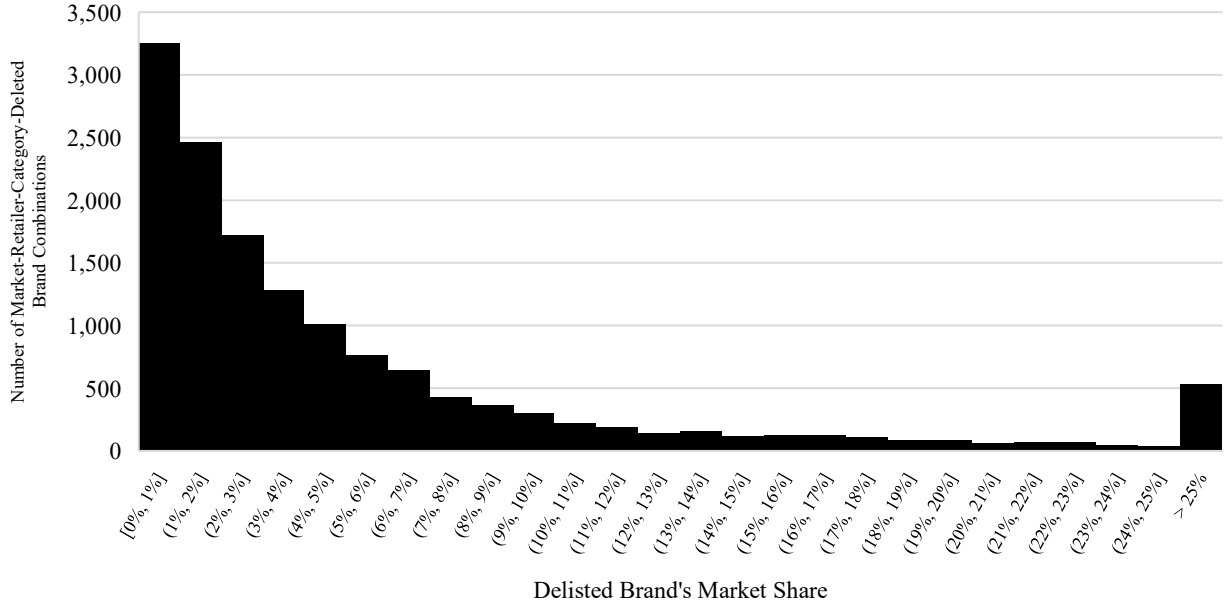


Table 2. Overview of Selection of Deleted Brands

				Pre-Deletion			Average Number of Competing Brands by Market-Retailer			Total Number of Competing Brands Across Markets and Retailers			
#	Brand	Category Name	Yearly Revenues in Million[a]	# Market-Retailer-Pairs	Market Share	Line Length	Avg. Length of Deletion in Weeks	# Rival national brands	# Sister Brands	# Private Labels	# Rival Brands	# Sister Brands	# Private Labels	∑
1	Pediasure	Complete Nutritional Products	$23.86	117	17.0%	4.51	11.09	.53	1.32	.27	62	154	32	248
2	Maalox	Antacids	$7.34	169	3.3%	5.99	21.47	3.28	.46	.92	554	78	155	787
3	Wisk	Detergents - Heavy Duty - Liquid	$7.17	120	2.0%	2.30	16.42	3.30	3.73	.39	396	447	47	890
4	Nature's Pride	Bakery - Bread - Fresh	$6.57	75	2.4%	6.90	7.63	3.68	.00	.59	276	0	44	320
5	Aunt	Frozen Waffles, Pancakes, French Toast	$5.57	110	4.2%	3.83	13.08	2.65	.00	.85	292	0	94	386
6	Truly	Alcoholic Cocktails	$5.15	35	24.4%	3.75	19.43	1.66	.00	.00	58	0	0	58
7	Kraft	Breakfast Bars	$4.69	120	6.5%	3.09	4.93	3.11	.00	.88	373	0	105	478
8	Soft 'N Gentle	Toilet Tissue	$4.17	76	3.6%	1.56	3.92	4.08	.00	.92	310	0	70	380
9	Digiorno	Cheese - Shredded	$4.14	95	1.1%	2.45	25.82	2.64	.00	.87	251	0	83	334
10	Kodak	Disk, Diskette, and Data Cartridge	$3.95	158	8.2%	3.18	1.41	2.30	.00	.00	364	0	0	364
…														
100	General	Coffee - Soluble Flavored	$1.09	45	19.4%	8.15	1.00	.91	.38	.16	41	17	7	65
101	Wonder	Fruit-Refrigerated	$1.09	n.d.	24.2%	7.09	39.33	1.00	.00	.00	3	0	0	3
102	Annie's	Yogurt-Refrigerated	$1.08	28	0.9%	3.87	5.18	7.96	.18	.54	223	5	15	243
…														
200	E.D.	Fresh Herbs	$.69	6	12.3%	1.82	10.33	.33	.00	1.00	2	0	6	8
201	Freezer	Entrees - Meat - 1 Food - Frozen	$.68	18	8.8%	3.96	11.17	4.89	.00	.83	88	0	15	103
202	Bertolli	Bakery - Dessert Cakes - Frozen	$.68	34	10.9%	2.64	30.12	1.88	.00	.24	64	0	8	72
…														
300	El Pollo Loco	Entrees - Mexican - 1 Food - Frozen	$.44	6	2.1%	2.35	20.33	5.50	.00	.17	33	0	1	34
301	Answer	Pregnancy Test Kits	$.44	16	5.0%	1.65	47.75	.69	.56	.88	11	9	14	34
302	Valio	Dairy-Milk-Refrigerated	$.44	5	0.3%	2.68	13.20	2.60	.00	.20	13	0	1	14
…														
400	Simply	Soup Mixes - Dry & Bases	$.27	19	2.0%	2.97	48.37	6.37	.00	.37	121	0	7	128
401	Valleydale	Frankfurters-Refrigerated	$.27	10	2.9%	2.43	2.30	3.60	1.70	.70	36	17	7	60
402	Dang	Coconut	$.27	5	32.5%	1.65	27.40	.00	.00	1.00	0	0	5	5
…														
500	Riggio	Fresh Vegetables-Remaining	$.18	n.d.	7.9%	7.56	11.00	4.50	.00	1.00	9	0	2	11
501	Gold	Hand Cleaners And Hand Sanitizers	$.18	17	5.1%	2.56	42.65	.94	.00	1.00	16	0	17	33
502	Healthy	Dry Dinners - Pasta	$.18	13	0.7%	4.26	35.46	6.38	.00	.85	83	0	11	94
…														
1,046	Toni&Guy	Creme Rinses & Conditioners	$.01	n.d.	0.1%	2.96	10.00	7.00	2.00	.00	7	2	0	9
		Mean =	$.45	13.83	6.3%	3.29	22.26	3.64	.22	.57	Sum =50,663	4,099	8,275	63,037

Notes: Total number of competing brands reflects the sum of competing brands in our sample across markets and retailers. We define deleted brands as having no sales and no distribution for at least one year. n.d. = not displayed; we do not show the number of market-retailer pairs if there are fewer than 5 to protect the retailer's identity.
[a] This is not the same as the total revenue in the U.S., but based on our sample frame, which is based on NielsenIQ's sample of stores.

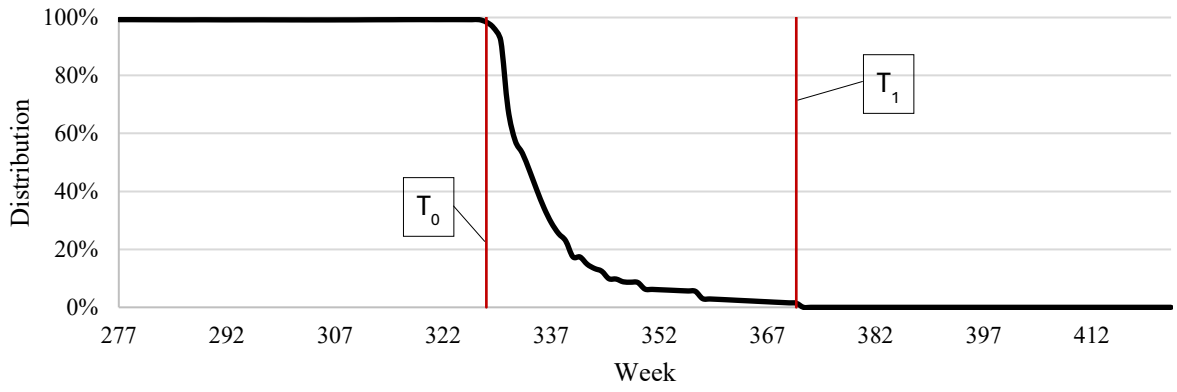
Brand deletions typically are not executed simultaneously across the entire U.S. at once but rather occur sequentially, separately by markets (DMAs) and within markets, by retailers. We account for the role of retailers as gatekeepers by analyzing brand deletions at the market-retailer-brand level (14,462 market-retailer-deleted brand combinations). This allows us to include retailer-specific marketing mix adjustments and retailer-specific moderators of the treatment effect.

Within each market and retailer, a brand deletion is not an instantaneous event. A brand deletion begins with a drop in its distribution (T_0) and ends when the brand is no longer distributed (T_1). Distribution is the weighted average of the percentage of stores that carry the

brand, where the weight is the store's total annual revenue (all commodity value; Farris, Olver, and De Kluyver 1989). To assess the beginning of the deletion, we smooth the distribution using a moving average over 13 weeks. We then define T_0 (beginning of the deletion) as the earlier of either the last increase in distribution (i.e., no subsequent increase) before the downward slide begins (98% of cases) or a cumulative drop in distribution that reaches at least 20% without recovery to the previous distribution level (2% of cases).

Figure 3 depicts brand deletions for a simulated sample case: Talenti pulling out of the frozen novelties category in one market at one retailer. Here, the beginning of the deletion period is determined using the last increase in distribution prior to a downward slide. Figure W1 in Web Appendix B shows the distribution of brand deletion periods across all brand deletions at the market-retailer level in our sample. The average brand deletion period is 22 weeks, while more than 30% are deleted within one week, and almost 90% within one year.

Figure 3. Brand Deletion of Talenti from Frozen Novelties in One Market at One Retailer



Notes: The black line shows the distribution over time. The first vertical red line marks the beginning of the deletion (T_0), and the second red line indicates the end of the deletion (T_1). The data are similar to cases in our sample but are simulated to protect retailers' identities and are not based on NielsenIQ's data.

In the analysis, we capture the brand deletion treatment as a continuous variable, ranging from 0 before T_0 when the deletion starts to 1 at time T_1 when the deletion is complete. In the transition period, the variable is the fraction of the deletion that has been completed, calculated

as $TreatPost_t = Treatment * \left(1 - \frac{Distribution_t}{Distribution_{T_0}}\right)$, where Treatment is an indicator. For example, if the pre-deletion distribution level is 100% and it drops to 50% after a few weeks and then down to 0%, the treatment variable $TreatPost_{pbcmrt}$ equals 0, .5, and 1, successively. The continuous nature of the treatment variable accounts for treatment effect changes during the deletion period.

Competing Brands

Our analysis focuses on how a brand deletion affects competing brands at the level of individual retailers within markets. We study all market-retailer-category-brand deletion combinations with a deletion period of up to 104 weeks at the market-retailer-category level. For a clean analysis, we require no overlapping brand deletions in the same category in the deletion period; if that is the case, the quasi-experiment is discarded. To ensure enough economic substance, we include, for each market where the brand deletion happens, the market's top 10 competing brands (i.e., rival national brands, sister brands, and private labels) with at least $100 revenue per week, market, and retailer on average, and the market's top 3 retailers. We study all available combinations of these top 10 brands at the top 3 retailers, noting that not every brand is sold at every retailer. To ensure enough observations for inferences, we require at least 52 weeks of sales data before and after the brand deletion (cf. Cleeren, van Heerde, and Dekimpe 2013).

Each competing brand may be a sister brand, which is a national brand owned by the same brand manufacturer, a rival national brand owned by another brand manufacturer, or a private label owned by the retailer. In total, we observe 6,031 unique competing brands, potentially affected by the 1,046 brand deletions across markets and retailers.[3] As our focal unit of analysis, we use market-retailer-category-brand combinations, reflecting the combination of a

[3] Across markets and retailers, we observe 14,462 market-retailer-category-deleted brand combinations. In 95% (= 13,723/14,462), a rival national brand is present, in 57% (=8,275/14,462) a private label, and in 19% (=2,769/14,462) a sister brand.

competing brand at a retailer within a market that potentially benefits from a brand deletion from that same market and retailer. Our analysis includes competitor marketing mix variables to account for changes in their marketing conduct prior to, during, and after the brand deletion. As importantly, these variables can be actively adjusted: for instance, manufacturer-driven decisions such as price promotions and feature advertising have frequently been shown to shift in response to competitive shifts. Also, retailer decisions with manufacturers' input, such as regular price and line length, can be altered; the latter by launching SKUs that already exist in other markets. Even a retailer-driven decision, such as distribution, though often high, is rarely 100% across all stores in a market (not even for private labels), leaving scope for further adjustment. We note that a retailer has, on average, 21.21 stores in a specific market.

To illustrate what market-retailer-category-brand pairs look like, let us consider the brand deletion by Talenti from the frozen novelties category in a simulated market at one retailer (Figure 2). Talenti is manufactured by Unilever, and the three sister brands available in the simulated market are Magnum, Klondike, and Popsicle. The six rival national brands are Blue Bunny, Dreyer's, Häagen-Dazs, M&M, Nestle, and The Skinny Cow, and there is one private label. Therefore, this brand deletion potentially affects 10 competing brands.

Variable Operationalization and Descriptive Statistics

Table 3 lists the operationalizations of the variables used in the study, including value sales (revenue), the marketing mix, and relevant control variables. We provide summary statistics by deleted brand and type of competing brand in Table W1 of Web Appendix B. Web Appendix C shows low to moderate correlations between the model variables, mitigating potential multicollinearity concerns.

Table 3. Variable Operationalizations

Variable	Operationalization (Supporting Reference)
Dependent Variable	
$Revenue_{\text{bcmrt}}$	Revenue (value sales) in $ for brand b in category c, market m, retailer r, and week t.
$CatRevenue_{\text{cmrt}}$	Revenue (value sales) in $ for category c, market m, retailer r, and week t.
Treatment*Post variable	
$TreatPost_{pcmrt}$	Treatment indicator variable (1 for treated units, 0 for control units) multiplied by the extent to which brand p has been deleted in category c, market m, retailer r, and week t, defined as $1 - Distribution_{pcmrt}/Distribution_{pcmrT_0} = Post_{pcmrt}$
Marketing Mix (Brand Level)	
Manufacturer-driven Decisions	
Promotion Price ($MKTG_{1,bcmrt}$)	SKU- and store-weighted[a] ratio of actual price to regular price for brand b in category c and in market m at retailer r in week t (Datta, Ailawadi, and van Heerde 2017); multiplied by 100.
Feature Advertising ($MKTG_{2,bcmrt}$)	SKU- and store-weighted[a] indicator variable for whether or not each available SKU of brand b in category c and in market m at retailer r has a feature advertising in week t (Datta et al. 2017); multiplied by 100.
Retailer Decisions with Manufacturers' Input	
Regular Price ($MKTG_{3,bcmrt}$)	SKU- and store-weighted[a] regular price of brand b in category c and market m at retailer r and week t, defined for each available SKU as the highest actual price (dollar revenue divided by units sold) per equivalent volume of the SKU in the most recent four weeks (Datta et al. 2017). Actual prices are deflated using the Consumer Price Index (van Heerde et al. 2013).
Line Length ($MKTG_{4,bcmrt}$)	Number of unique SKUs sold by brand b in category c and in market m at retailer r in week t (Datta et al. 2022).
Retailer Decisions	
Distribution ($MKTG_{5,bcmrt}$)	Store-weighted[b] distribution of brand b in category c and in market m and retailer r and week t; multiplied by 100.
Control Variables (Controls)	
Competitors' Promotion Price	Market share–weighted average of brand b's competing brands' (≠b) promotion price in category c, market m at retailer r and week t (Datta et al. 2022).
Competitors' Regular Price	Market share–weighted average of brand b's competing brands' (≠b) regular price in category c, market m at retailer r and week t (Datta et al. 2022).
Marketing Mix (Category Level)	
$CategoryPrice_{cmrt}$	Average price in category c, market m, retailer r and week t, calculated as category revenue divided by volume sold.
$CategoryFeature_{cmrt}$	SKU- and store-weighted[a] indicator variable for whether or not each available SKU in category c in market m at retailer r has a feature advertising in week t.
$CategoryLineLength_{cmrt}$	Number of unique SKUs sold in category c in market m at retailer r in week t.
Moderators	
$SB\ present_{pcmr}$	Indicator variable (=1) for whether a sister brand of deleted brand p is present in category c, market m, at retailer r; 0 else.
$PL\ present_{cmr}$	Indicator variable (=1) for whether a private label is present in category c, market m, at retailer r; 0 else.
$PriceSimilarity_{pbcmr}$	1 – relative price gap of competing brand b in category c, market m, retailer r, in the year prior to the start of brand deletion (t=T_0). Relative price gap between competing brand b and delisted brand p is defined as: $\frac{(\lvert MKTG_{3,bcmrt} - MKTG_{3,pcmrt} \rvert)}{(\lvert MKTG^{MAX}_{3,cmrt} - MKTG^{MIN}_{3,cmrt} \rvert)}$. $MKTG_{3,bcmrt}$ ($MKTG_{3,pcmrt}$) is the regular price of competing brand b (deleted brand p). $MKTG^{MAX}_{3,cmrt}$ ($MKTG^{MIN}_{3,cmrt}$) is the maximum (minimum) regular price in category c, market m, retailer r in week t.
$DeletedBrandShare_{pcmr}$	Revenue-based market share of the deleted brand p in category c, market m, retailer r, in the year prior to the start of brand deletion (t=T_0).
$CompetitorBrandShare_{bcmr}$	Revenue-based market share of the competing brand b in category c, market m, retailer r, in the year prior to the start of brand deletion (t=T_0).
$CategoryConcentration_{pcmr}$	Concentration (Hirschman-Herfindal Index) of category c in market m and retailer r in the year prior to the start of brand p's deletion (t=T_0).
$CategoryGrowth_{pcmr}$	Growth (% year-on-year change in) revenue of category c in market m and retailer r at the start of brand p's deletion (t=T_0), measured as the average first difference of log category (value) sales over the 52 weeks before T_0.

[a] To aggregate SKU-level metrics to brand-level metrics, we compute weighted averages for the focal variables across a brand's SKUs within each store that stocks the brand, using the SKU's share of brand sales in the store as weights, for a rolling window of the previous 13 weeks. Then we aggregate across all stocking stores, with weights equal to each store's share of total category sales in the same rolling window (Datta et al. 2017). We focus on all SKUs (after ordering them from largest to smallest sales) that collectively account for 95% of a brand's sales within a calendar year.

[b] Weights are equal to each store's share of total category sales in market m in a rolling window of the previous quarter. We use these weights to calculate retailer-level distribution, which is the aggregate of store-level distribution where a brand is either present (1) or not (0).

METHOD

To estimate the effect of a brand deletion on competing brand revenue and marketing activity, our analysis proceeds as follows. We leverage a before–after–with–control setting to estimate the treatment effect of a brand deletion on competing brands, at the market-retailer-category-competing brand level (63,037 combinations). We use a difference-in-differences model applied to data at the market (m)-retailer (r)-category (c)-competing brand (b)-week (t) level. The model is augmented with moderators of the treatment effect and marketing mix effects. We also estimate difference-in-differences models for the levels of the marketing mix variables of competing brands, to assess how much they engage in marketing conduct changes.

We now discuss some identification challenges for estimating treatment effects and how we address them. First, competing brands' sales can change over time, even without the brand deletion. To account for this possibility, we use as a control the *same* competing brand at the *same* retailer in other markets. This approach accounts for seasonal effects and trends that affect both the treated and control units. The fact that we observe "staggered treatment timing" for each brand deletion and expect heterogeneous treatment effects requires special care because the standard two-way fixed effects estimator may be biased (Goodman-Bacon 2021). We deal with this as follows: we identify for each "treated" case (a market-retailer-category-deleted brand-competing brand combination) a set of control units that consist of "not-yet-treated" cases. That is, these "not-yet-treated" cases are not treated while they serve as a control unit, i.e., at least until one year after the end of the post-deletion period, and as such, they present clean control cases. We then stack all treated-control sets and estimate a difference-in-difference model with extensive fixed effects for cross sections and time periods. Web Appendix D presents an illustrative example of the data structure, highlighting that control units are never treated while

they serve as controls. Moreover, as we highlight in the Robustness Check section, we also estimate an alternative two-stage approach (Gardner et al. 2024) that explicitly takes the staggered treatment timing into account, and find very comparable results.

Second, the timing of a brand manufacturer's deletion decision for a given market and retailer is unlikely to be random, and competing brands may strategically employ their marketing mix to improve the share of revenue they gain after a brand deletion. We correct for the endogeneity of the deletion decision and marketing mix using a control function approach with Hausman-style instruments, explained below. Third, the deletion effect may be dynamic rather than constant over time, as it may take a few weeks or months before the deletion is complete, which we accommodate through our continuous *TreatPost* variable defined earlier.

Treated and Control Units

The treated units are competing brands sold at retailers in markets where the focal brand is transitioning from being available to being deleted. To create a comparable control unit (i.e., where the focal brand is not deleted yet) for each treated unit, we select the possible controls to match the treated units. We use as a control the same competing brand as the treated competing brand in the same time period at the same retailer, but in different markets that still feature the (not yet) deleted brand. For example, if competing brand b at retailer r in market 1 is potentially benefiting from the deletion of brand p, we use brand b by retailer r in markets 2, 3, and 4 as control units if brand p is still active in these markets. By using the same brand b and retailer r, we ensure the treated and control brands are identical in general brand characteristics (e.g., publicity, national brand advertising) and retailer characteristics (e.g., fit of brand with retailer positioning, strength of retailer). By design, this implies that the market(s) in which a brand b at retailer r is deleted last will not have control unit.

Thus, potential control units are the same competing brand, sold at the same retailer, operating in all markets where the focal brand still operates. To find the most similar control units, we match based on log weekly revenue and log marketing mix in the year prior to the start of the deletion at time T_0. Matching on the pretreatment outcome (log revenue) entails matching on unobservable brand–market characteristics that relate to the outcome measure itself (Abadie 2021), which is desirable because it means the analysis accounts for unobservables.[4] For each potential control, we calculate the correlations between the treated unit's and potential control unit's weekly log revenue and five log marketing mix variables across these 52 weeks. To ensure parallel trends pretreatment, we select for each treated unit up to 10 control units (in line with, e.g., Bollinger and Sexton 2023, p. 292; Keller and Guyt 2025, pp. 11-12) that each must have an average correlation of at least .8 with the treated unit across these six variables. A correlation of .8 is typically considered high, and we note that this threshold already leads to a high average pretreatment correlation of > .89 (Table 4) between treated and control units. However, we acknowledge that it is an arbitrary threshold and assess the robustness of our findings to even higher thresholds of .85 and .90, respectively (see Additional Checks section).

Table 4. Pretreatment Correlations between Treated and Control Units by Brand Type

Variable	Overall	Rival National Brand	Sister Brand	Private Label
Revenue	.80	.80	.82	.80
Promotion Price	.93	.94	.94	.90
Feature Advertising	.96	.96	.96	.95
Regular Price	.91	.91	.91	.91
Line Length	.91	.91	.90	.92
Distribution	.87	.87	.87	.87
Average	**.89**	**.89**	**.90**	**.89**

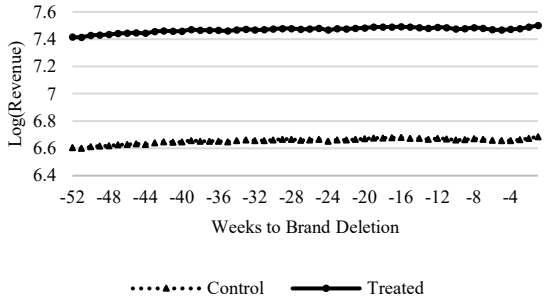
[4] It is possible to consider other variables in defining control units. However, our approach is to define control markets as similar to the treatment market by considering how similar revealed brand preferences (revenue) and revealed marketing mix activity (marketing variables) are between markets. Underlying these similarities (e.g. preferences for budget brands) may be certain demographics (e.g., low wages), but we believe it is more direct to use observed preferences and marketing mix conduct to find control units similar to the treated unit.

Parallel Trends

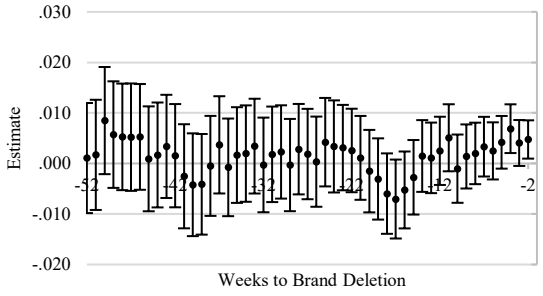
A key assumption of the difference-in-differences model is that the treated unit's outcome would have been the control unit's outcome in the absence of the treatment. We cannot test the posttreatment parallel trends assumption, because the counterfactual outcome is not observed, but we can assess whether treated and control units evolve in parallel prior to the treatment. Figure 4a shows visual evidence of parallel trends for revenue, with an average pretreatment correlation of .998. Figure 4b shows the pretreatment deviation test (where we control for cross-section and time fixed effects). For 50 [48] out of 52 weeks, there is no statistical difference (p>.05) [p>.1], which is close to what is expected by chance.[5] We also conduct an in-time placebo test, where we set the treatment date 26 weeks before the actual treatment. The difference is negligible (b = -.001, p=.837), confirming pretreatment parallel trends.

Figure 4. Evidence for Pretreatment Parallel Trends

Panel a: Parallel Pretreatment Trends for Treated and Control Units (r = .998)



Panel b: Pretreatment Deviation Test for Treated vs. Control Units



Notes: Panel A shows the aggregate evolution of revenue before the beginning of the deletion (T_0), separately for treated and control units. Web Appendix E shows the evolution by outcome variable at a more disaggregate level. Panel B shows the parameter estimates of the pretreatment deviation test, where each dot is an estimate representing the difference between treated and control units, with the estimate of the immediate week before T_0 (-1) serving as baseline. Bars show 95% confidence intervals. We provide detailed parameter estimates in Web Appendix F.

Model-Free Evidence of Brand Deletion Effects on Competing Brands

To provide first insights into how brand deletions affect competing brand sales and

[5] The four lowest p-values are .005, .014, .077, and .080, respectively.

marketing conduct, Table 5 presents the average relative change in revenue and the marketing mix before the deletion started (52 weeks before), and 52 weeks after the deletion finished. Table 5 is based on averaging (63,037) case-specific values, comparing for each case the before-after differences of treated and (average) control units. We do this separately for rival national brands, sister brands, and private labels.

Table 5. Model-free Evidence on Changes in Outcome Metrics

	Outcome	Manufacturer-driven Decisions		Retailer Decision with Manufacturers' Input		Retailer Decisions
	Revenue	**Promotion Price**	**Feature Advertising**	**Regular Price**	**Line Length**	**Distribution**
Rival National Brand	1.57%	-.01%	5.05%	.07%	.35%	.09%
Sister Brand	1.19%	-.03%	4.60%	.12%	.65%	.10%
Private Label	1.44%	.01%	16.30%	.20%	.38%	.03%
Overall	**1.52%**	**.00%**	**6.08%**	**.09%**	**.37%**	**.08%**

Notes: To minimize the impact of outliers at the case-level, we trim the case-specific effects at the 1st/99th level.

Table 5 shows that revenue changes notably for all brand types. After the deletion, all three brand types gain sales, at between 1.19% and 1.57% of their before-deletion levels. In terms of marketing variables, we find that while promotion prices hardly change, feature advertising, regular price, line length, and distribution generally increase following the deletion. Noting the usual caveats regarding model-free comparisons, we turn to our formal analysis to provide cleaner estimates of brand deletion effects.

Difference-in-Differences Model for Brand Revenue

As discussed, the treatment and control pairs meet the pretreatment parallel-trends assumption. Following Li and Van den Bulte (2023, p. 751), "if the testable part of the DID parallel trends assumption holds, then use the DID method," we adopt a moderated difference-in-differences model. We stack weekly observations for all treated units and all control units (up to 10 per treated unit), for a total of 55,876,309 observations across all market-retailer-category-

brand-week combinations. We estimate a moderated difference-in-differences model for the effect of brand deletion on competing brand revenue, accounting for moderators, marketing mix effects and changes therein, control variables, and endogeneity control functions:

$$Revenue_{bcmrt} = \underbrace{\alpha_{bcmr} + \alpha_t}_{\text{fixed effects}} + \underbrace{\beta_{comp} TreatPost_{pcmrt}}_{\text{level shifts by brand type}} + \underbrace{\left(\eta_{1,comp}^{\Delta post,ctrl} SB\ present_{pcmr} + \eta_{2,comp}^{\Delta post,ctrl} PL\ present_{cmr}\right) \times Post_{pcmrt}}_{\text{pre vs.post difference in competing brand presence effects}} \quad (1)$$

$$+ \underbrace{\left(\eta_{1,comp}^{\Delta post,\Delta treat} SB\ present_{pcmr} + \eta_{2,comp}^{\Delta post,\Delta treat} PL\ present_{cmr}\right) \times TreatPost_{pcmrt}}_{\text{effect of competing brand presence on treatment effect}} + \underbrace{\delta_{comp}\ Post_{pcmrt}}_{\text{pre vs.post changes by brand type}}$$

$$+ \sum_{q=1}^{5} \left\{ \underbrace{\theta_q^{\Delta post,ctrl} MOD_{q,pbcmr} \times Post_{pcmrt}}_{\text{pre vs.post differences in moderator effects}} + \underbrace{\theta_q^{\Delta post,\Delta treat} MOD_{q,pbcmr} \times TreatPost_{pcmrt}}_{\text{moderators' effect on treatment effect}} \right\} + \underbrace{\phi CF_TreatPost_{pcmrt}}_{\text{control function for brand deletion}}$$

$$+ \sum_{k=1}^{5} \left\{ \underbrace{\gamma_{comp,k} MKTG_{k,bcmrt}}_{\text{marketing mix effects}} \right\} + \sum_{k=1}^{5} \left\{ \underbrace{\varphi_k CF_{MKTG\,k,bcmrt}}_{\substack{\text{control functions} \\ \text{marketing mix}}} \right\} + \sum_{j=1}^{2} \underbrace{\delta_j Controls}_{\text{competitor price and promotion}} + \varepsilon_{bcmrt}$$

The fixed effects α_{bcmr} in Equation 1 account for unobserved level differences in brand revenue specific to brand b in category c in market m and at retailer r (the cross section). The fixed effects α_t account for any time-varying unobservables such as trends. The first effect of interest is $\beta_{1,comp}$, which is the effect of brand p's deletion on revenue the competing brand ($TreatPost_{pcmrt}$),where we distinguish between three brand types: comp ={rival national brand, sister brand, private label}. Equation 1 also controls for brand type-specific pre versus post changes in sales through $Post_{pcmrt}$.[6]

$SB\ present_{pcmr}$ indicates whether a sister brand of the deleted brand p is active in the category at that retailer and market. Similarly, $PL\ present_{cmr}$ reflects whether a private label is present. $MOD_{q,pbcmr}$ denotes moderator q (defined in Table 3), and $MKTG_{k,pbcmrt}$ captures marketing mix variable k (defined in Table 3: 1 = promotion price, 2 = feature advertising, 3 =

[6] $Post_{pcmrt}$ has a subscript p to indicate that the same market-retailer-category-competing brand combination may appear multiple times in the data. This occurs because it can serve as control for different treated cases and, if the brand deletion timing differs, the $Post_{pcmrt}$ variable may take on different details. As such, subscript p technically has its own subscripts for category (c), market (m), and retailer (r) because a deletion of brand p happens at the market-retailer-category level. We omit these for readability reasons.

regular price, 4 = line length, 5 = distribution,). $CF_TreatPost_{pcmrt}$ and $CF_MKTG_{k,bcmrt}$ are control function terms addressing potential endogeneity of the brand deletion decision and the marketing mix, respectively. $Controls$ includes the weighted average competitor regular price (j=1) and promotion price (j=2), as defined in Table 3. All marketing mix variables are mean-centered within a cross section, and all moderators are mean-centered (as they are constant within a cross-section). The main effects of $Treat_{pcmr}$ and its interaction effects with (the time-invariant) $MOD_{q,pbcmr}$ variables are subsumed in the fixed effects α_{bcmr}.

Identification Challenges

The treated versus control unit approach accounts for unobservables that could drive the revenue change, but we note other potential threats to identification that require attention, including the timing of the brand deletion and the endogenous nature of marketing mix tools.

Endogeneity of the timing of brand deletion

The order in which a brand manufacturer deletes a brand from different markets and retailers may not be random but instead could be determined by the strength of the remaining sister brands in these markets or other unobserved factors. Endogenous timing of the brand deletion would be reflected in the $TreatPost_{pcmrt}$ variable, which starts to take on non-zero values once the brand deletion begins. Unobserved revenue confounders may be correlated with this variable, in which case the parameter estimates of Equation 1 may be inconsistent (Wooldridge 2010). To address this concern, we specify a first-stage model using exogenous variation to explain the extent to which brand p is deleted from competing brand b's category c in market m at retailer r in week t, as reflected in $TreatPost_{pcmrt}$. From this model, we derive the residual $CF_TreatPost_{pcmrt}$ and include it in Equation 1 to alleviate the endogeneity concern.

To model $TreatPost_{pcmrt}$, we include an instrumental variable. This instrument must meet the criteria of strength, validity, and granularity (Shi, Grewal, and Sridhar 2021), i.e., it should (1) correlate with the outcome variable (whether a brand is deleted), (2) not affect the unobserved parts of competing brands' revenues, and (3) vary at the same level as the data.

Following Sridhar et al. (2016) and Keller and Guyt (2023), we leverage institutional insights to determine an appropriate instrument, namely, the share of neighbors-of-neighboring markets in which the brand has already been deleted at the retailer. A market (DMA) has neighboring markets, which in turn have their own neighbors, which we define as neighbors of neighbors. For example, the Lafayette, LA, market has three neighbors (Lake Charles, LA, Alexandria, LA, and Baton Rouge, LA). Lake Charles, LA, has one neighbor that is not a neighbor of Lafayette, LA (Beaumont-Port Arthur), Alexandria, LA, has two (Shreveport and Monroe-El Dorado), and Baton Rouge's neighbors include New Orleans, LA, and Jackson, MS. Together, these 5 markets are Lafayette's neighbors' neighbors. If two out of these five neighbors have deleted the brand already, the instrument takes the value .4 (=2/5).

The rationale for the instrument is as follows. Brands are typically managed and distributed by geographic regions covering multiple markets. The decision to delete a brand from a region is usually made for all markets in that region, regardless of individual market performance. Even if a single market represents a success, it may not be enough to support the overall overhead costs of distributing and managing the entire region. Because contracts and inventory levels vary across retailers and markets, the brand does not disappear simultaneously but sequentially, so brand deletions in neighbors-of-neighboring markets could be informative about whether a deletion is imminent for the focal market (leading to instrument strength; see also Hwang, Bronnenberg, and Thomadsen (2010) and Keller and Guyt (2023)). Whether a brand

is deleted in neighbors-of-neighboring markets (hundreds of miles away) should not affect demand for that brand in the focal market, as consumers do not travel hundreds of miles for grocery shopping (creating instrument validity, i.e., meeting the exclusion restriction). We focus on neighbors' neighbors (rather than direct neighbors) to avoid potential cross-market demand spillovers. The share of neighbors' neighbors where the brand has already been delisted at a given retailer varies at the same level as our data, so this instrumental variable also meets the granularity criterion. Finally, we note that the instrument in the first-stage regression is positively and strongly related to the brand deletion decision ($b = .513$, $p < .001$); see Web Appendix G.

Endogeneity of the marketing mix

Brand managers may be strategic in setting their marketing mix, and those decisions can be driven by unobserved demand shocks that vary cross-sectionally or over time. Cross-sectional shocks reflect unobserved differences across market-retailer-brand combinations, such that a brand may be more popular in some markets and therefore command a price premium. To safeguard against this potential endogeneity concern, we only use within-market-retailer-category-brand variation by including market-retailer-category-brand fixed effects α_{bcmr}. The unobserved brand factors could also be time-varying (see, e.g., Papies, Ebbes, and van Heerde 2017), but the quasi-experimental method and chosen control units implicitly account for time-varying brand-specific factors' effect on the treatment effect (e.g., national brand advertising), because the control units pertain to the same brand at the same time.

What remains are time-varying factors specific to a given market-retailer-category-brand pair, as when brands strategically adjust their marketing mix response locally at a given market or retailer. To address this source of endogeneity, we use a control function approach with Hausman-style instruments. For example, for regular price, the instrumental variable is the

average regular price of the same brand b at the same retailer in neighbors' neighboring markets in the same week. In the first stage, we regress all marketing mix variables on the instrumental variables and control variables, and include the residuals ($CF_MKTG_{k,bcmrt}$) in Equation 1.

The rationale for instrument relevance for regular price is that underlying cost factors for ingredients, packaging, transportation, and labor may exhibit regional and longitudinal variation, affecting regular prices both in the focal market and its neighbors' neighbors (see also Ma et al. (2011) and Keller and Guyt (2023)). The identifying assumption is that markets operate in comparable economic environments and face similar cost shocks (e.g., due to price fluctuations or transportation costs), which influence marketing mix decisions. Thus, changes in the marketing mix at the same retailer in markets that are neighbors' neighbors should correlate with changes in the marketing mix for the retailer in focal markets, implying that the instrument is relevant and strong. Indeed, we find that the instruments in the first-stage regressions are all positively related to the endogenous variables (parameters range from .722 to .942, all $p < .001$), attesting to the instruments' strength. Multivariate Sanderson-Windmeijer tests also attest to the strength of each instrument (all $p < .001$); see Web Appendix G for details.

To satisfy the exclusion restriction, no common demand shocks should affect both markets, which is why we use neighbors' neighbors markets, as before. Because we aggregate across all neighbors' neighboring markets, it is unlikely that the instrument is affected by market-level idiosyncratic shocks (Sridhar et al. 2016). Therefore, we argue that the instruments are valid and satisfy the exclusion restriction. Finally, instruments need to exhibit sufficient granularity (Shi et al. 2021). In our case, they do; the average marketing mix across neighbors' neighboring markets in a given week varies at the same level (across markets, retailers,

categories, brands, and time) as our focal marketing mix variable.[7]

We estimate Equation 1 (and 2 below) with OLS using clustered errors at the retailer-category-brand level to account for potential autocorrelation and correlation across markets. Following Papies et al. (2017), we adjust the standard errors to reflect that the control function terms are estimated quantities.

Difference-in-Differences Model for Marketing Mix

To assess the extent to which competing brands actively change their marketing support in the wake of the brand deletion, we use the same quasi-experimental approach as for brand revenue. Stacking the same observations (where the same treated units are paired with the same up to 10 control units), we now estimate a difference-in-differences model for each marketing variable (k=1,…,5) separately:

$$MKTG_{k,bcmrt} = \underbrace{\alpha_{k,bcmr} + \alpha_{k,t}}_{\text{fixed effects}} + \underbrace{\beta_{k,comp} TreatPost_{pcmrt}}_{\text{level shifts by brand type}} + \underbrace{\left(\eta_{k1,comp}^{\Delta post,ctrl} SB\ present_{pcmr} + \eta_{k2,comp}^{\Delta post,ctrl} PL\ present_{cmr}\right) \times Post_{pcmrt}}_{\text{pre vs.post difference in competing brand presence effects}} \quad (2)$$
$$+ \underbrace{\left(\eta_{k1,comp}^{\Delta post,\Delta treat} SB\ present_{pcmr} + \eta_{k2,comp}^{\Delta post,\Delta treat} PL\ present_{cmr}\right) \times TreatPost_{pcmrt}}_{\text{effect of competing brand presence on treatment effect}} + \underbrace{\delta_{k,comp}\ Post_{pcmrt}}_{\text{pre vs.post changes by brand type}}$$
$$+ \sum_{q=1}^{5} \left\{ \underbrace{\theta_{kq}^{\Delta post,ctrl} MOD_{q,pbcmr} \times Post_{pcmrt}}_{\text{accounts for pre vs.post differences in moderator effects}} + \underbrace{\theta_{kq}^{\Delta post,\Delta treat} MOD_{q,pbcmr} \times TreatPost_{pcmrt}}_{\text{moderators' effect on treatment effect}} \right\} + \underbrace{\phi_k CF_TreatPost_{pcmrt}}_{\text{control function for brand deletion}} + \varepsilon_{k,bcmrt}$$

We note that the *TreatPost* variable is based on the distribution of the deleted brand p (which goes to zero eventually), while one of the marketing variables is the distribution of competing brand b. This separation ensures there is no circularity in the model.

Difference-in-Differences Model for Category Revenue

To assess what happens with category revenue in the wake of the brand deletion, we use a similar quasi-experimental approach as for brand revenue. After identifying up to 10 control units for each treated unit, we specify a difference-in-differences model for category revenue

[7] We note that peer groups need to be partially overlapping (Angrist 2014; Shi et al. 2021), as is true in our study setting, because we use the same brand sold at the same retailer in different markets and because we only include markets where the brand is still active (resulting in different peer groups within brands).

(method details in Web Appendix H, variable operationalization details in Table 4):[8]

$$CatRevenue_{cmrt} = \underbrace{\alpha_{cmr} + \alpha_t}_{\text{fixed effects}} + \underbrace{\beta_{c,comp} TreatPost_{pcmrt}}_{\text{level shifts by brand type}} + \underbrace{\gamma_1\ CategoryPrice_{cmrt}}_{\text{average category price}} + \underbrace{\gamma_2\ CategoryFeature_{cmrt}}_{\text{average category feature advertising}} \quad (3)$$

$$+ \underbrace{\gamma_3\ CategoryLineLength_{cmrt}}_{\text{total \# uniqe SKUs in category}} + + \underbrace{\delta_{c,comp}\ Post_{pcmrt}}_{\text{pre vs.post changes by brand type}} + \underbrace{\phi_c CF_TreatPost_{pcmrt}}_{\text{control function for brand deletion}} + \varepsilon_{pcmrt}$$

RESULTS

Difference-in-Differences Model Results for Brand Revenue

To understand the effect of brand deletion on competing brand revenue, we turn to the corresponding difference-in-differences model (Equation 1), the results of which are in Table 6. The model fits the data well (R^2 = .848) and collinearity does not appear to be an issue, as also confirmed in Web Appendix J, where we incrementally add blocks of variables to the model, starting with just intercept shifts. Web Appendix C also provides a correlation matrix among the key variables. Most of the control function terms are significant at the 10% level, indicating the relevance of accounting for the endogeneity of marketing conduct and the delisting decision. We do note that Web Appendix J shows that directionally, the findings are not different when we leave out the control function terms.

Results for Brand Deletion Effects on Competing Brand Revenues

Table 6 shows that the effect of brand deletion is generally positive, but not significant for rival national brands (b=.004; *p*=.661) and sister brands (b=.009; *p*=.588), while private labels gain revenue (b=.021, *p*=.035)[9]. These effects hold for moderators and marketing mix kept at their means, as all covariates are mean-centered. However, as discussed below in Table 9,

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[8] In an auxiliary analysis, we also measure the impact of a brand deletion on whether retailers launch new brands in the category. Specifically, we assess whether the rate of new brands (measured in three ways: brands (i) new to the market-retailer pair, (ii) new to the market, and (iii) new to the category) differs between market-retailer-category combinations with a brand deletion versus its control markets. Using a difference-in-difference specification, we find that the rate of launching new brands is not different depending on whether a brand deletion takes place or not. Please see Web Appendix I for details.

[9] To arrive at percentage changes, we transform the intercept shifts by (e^b-1)*100%.

when we take the moderators and marketing mix conduct into account, the insights change.

Table 6. Effect of Brand Deletion, Moderators, Marketing Mix on Competitor Revenue

Variable	Estimate	Std. Error	p-value
Intercept shifts			
Treat x Post x …			
Rival National Brand	.004	.009	.664
Sister Brand	.009	.016	.588
Private Label	.021	.010	.035
Post x …			
Rival National Brand	-.012	.006	.034
Sister Brand	-.016	.015	.271
Private Label	-.030	.010	.004
Moderators			
Treat x Post x …			
Sister Brand Present x Rival National Brand	-.004	.006	.495
Sister Brand Present x Private Label	.017	.012	.171
Private Label Present x Rival National Brand	.003	.007	.638
Private Label Present x Sister Brand	-.010	.021	.631
Price Similarity	.028	.011	.010
Deleted Brand Share	.246	.074	.001
Competing Brand Share	.034	.021	.106
Category Concentration	-.158	.031	.000
Category Growth	-.018	.456	.968
Marketing Mix			
Rival National Brand x …			
Promotion Price	-1.765	.019	.000
Feature Advertising	.034	.000	.000
Regular Price	-.049	.024	.046
Line Length	1.266	.025	.000
Distribution	2.942	.065	.000
Sister Brand x …			
Promotion Price	-1.685	.038	.000
Feature Advertising	.037	.001	.000
Regular Price	-.256	.104	.014
Line Length	1.325	.107	.000
Distribution	3.098	.163	.000
Private Label x …			
Promotion Price	-1.591	.080	.000
Feature Advertising	.021	.001	.000
Regular Price	-.179	.026	.000
Line Length	1.190	.044	.000
Distribution	3.586	.227	.000
Competitor Marketing Mix	✓		
Post x Moderators	✓		
Control Function Terms	✓		
Cross-section FE	✓		
Time FE	✓		

Notes: Clustered standard errors by retailer-category-deleted brand-competing brand combinations.
N = 55,876,309

Table 6 shows that competing brands gain more revenue when they are more similarly priced to the deleted brand (b=.028, *p*=.010), indicating consumers may switch to similarly-

priced competitors rather than brands with a different price positioning. Competing brands also gain when the deleted brand had a higher market share (b=.246, p<.001), which makes sense as there is more demand to redistribute. A higher category concentration limits the gain for competing brands (b=-.158, p<.001), as consumers may find it harder to find a substitute in a category with fewer yet bigger brands. The market share of the competitor has a marginally significant effect on revenue (b=.034, p=.106), indicating that larger brands may be better positioned to scoop up revenue from the deleted brand. The category growth does not materially affect the revenue of competing brands.

Results for Marketing Mix Effectiveness

Both the dependent variable and the marketing mix covariates are in log form, and the parameters can be interpreted as elasticities with respect to revenue. The parameter estimates for the marketing mix are face valid. The promotional price elasticities (stronger in magnitude than regular price elasticities, as expected; Bijmolt, van Heerde, and Pieters 2005) are negative for rival national brands (b=-1.765, p<.001), sister brands (b=-1.685, p<.001), and for private labels (b=-1.591, p<.001). The parameter for feature advertising is also an elasticity (percentage change in revenue due to a 1% increase in feature support), as feature support is defined as the log of the weighted fraction of stores and products having feature advertising for the brand in that week. Feature advertising has positive yet weak elasticities on brand revenue for all types of brands: rival national brands (b=.034, p< .001), sister brands (b=.037, p<.001), and for private labels (b=.021, p<.001). Regular price elasticities for brand revenue are negative for rival national brands (b=-.049, p=.046), sister brands (b=-.256, p=.014), and for private labels (b=-.179, p<.001), suggesting price-elastic brand demand in unit sales. Line length elasticities are positive and strong in magnitude for rival national brands (b=1.266, p<.001), sister brands (b=1.325,

p<.001), and (at a lower magnitude) for private labels (b=1.190, *p*<.001).[10] We find very strong positive distribution elasticities for rival national brands (b=2.942, *p*<.001), sister brands (b=3.098, *p*<.001), and for private labels (b=3.586, *p*<.001). We note that these distribution elasticities are based on within-retailer, within-market time series variation and therefore not directly comparable to more aggregate distribution elasticities based on national-level time series variation.[11]

Results for Brand Deletion Effect on Competitor Marketing Conduct

We now discuss competing brands' marketing mix changes after a brand deletion.[12] Table 7 shows the results from Equation 2. Focusing on the focal treatment x post parameters, i.e., the relative change in marketing conduct of the treated markets to the control markets, we find that (for moderators at their means) rival national brands offer slightly deeper promotion price discounts (b=-.001, *p*=.043), as do competing brands that are larger (b=-.002, *p*=.001), possibly indicating more sizeable resources to react. Store brands offer deeper promotions if private labels are present (b=-.003, *p*=.007), as do private labels, if store brands are present (b=-.001, *p*=.003). These findings suggest that, especially when store brands and private labels are both present, they strongly compete through price promotion in the wake of the brand deletion.

Conversely, price promotions are shallower for brands that have lower price similarity to

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[10] In a robustness check, we also added a brand's fraction of SKUs that are new to the category since the beginning of the brand deletion as an additional moderator (interacted with TreatPost and Post). The interaction effect with TreatPost is economically small (b=.043, *p*=.386) while the focal parameters are not affected in sign or magnitude. We present these results in Web Appendix K.

[11] Web Appendix L discusses the magnitude of the distribution elasticities in more depth, including a transformation to (much lower) national distribution elasticities.

[12] We report on the change relative to the brand's control markets ($\beta_{k,comp}$). We could also consider the overall (before-after) change accounting for the control group's change ($\delta_{k,comp}$), i.e., $\beta_{k,comp} + \delta_{k,comp}$. For rival national brands, the overall promotion intensity shows no change (-.0006 + .0004 = -.0002, *p*=.572), while it becomes shallower for sister brands (i.e., at the same rate as the control group; .0017, *p*=.047). Rival national brand's feature advertising decreases at a lower rate than in the control group (-.0528, *p*<.001). Finally, line length decreases for rival national brands at the same rate as control units (-.0111, *p*=.003) and line length increases at the same rate as control units for private labels (.0132, *p*=.011).

the deleted brand (b=.001, p<.001), possibly reflecting that the deleted brand served customers in a different price segment. Price promotions are also shallower the larger the deleted brand was (b=.005, p=.012), reflecting reduced competitive pressure.

Table 7. Results for Effect of Brand Deletion on Competitor Marketing Conduct

	Manufacturer-driven Decisions						**Retailer Decisions with Manufacturers' Input**						**Retailer Decisions**		
	Promotion Price			**Feature Advertising**			**Regular Price**			**Line Length**			**Distribution**		
Variable	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.
Intercept shifts															
Treat x Post x …															
Rival National Brand	-.001	.000	.043	.022	.009	.013	.024	.003	.000	.001	.003	.695	.006	.001	.000
Sister Brand	.000	.001	.847	-.016	.021	.443	.033	.004	.000	.009	.007	.179	.011	.003	.000
Private Label	.000	.000	.218	.048	.012	.000	.012	.003	.000	-.006	.004	.135	.006	.001	.000
Post x …															
Rival National Brand	.000	.000	.049	-.075	.006	.000	.007	.001	.000	-.012	.002	.000	.000	.001	.587
Sister Brand	.002	.000	.001	-.001	.015	.940	-.012	.004	.003	-.007	.006	.260	-.006	.002	.024
Private Label	.000	.000	.536	.086	.013	.000	.051	.005	.000	.019	.005	.000	.003	.001	.000
Moderators															
Treat x Post x …															
Sister Brand Present x Rival National Brand	.000	.000	.597	-.008	.007	.225	.002	.002	.167	-.002	.002	.256	-.001	.001	.249
Sister Brand Present x Private Label	-.001	.000	.003	-.049	.020	.013	.009	.005	.075	.003	.005	.604	-.005	.003	.079
Private Label Present x Rival National Brand	.000	.000	.818	-.003	.006	.619	.000	.002	.864	-.007	.002	.000	-.002	.001	.054
Private Label Present x Sister Brand	-.003	.001	.007	.058	.024	.015	-.007	.004	.051	-.003	.008	.734	-.009	.005	.055
Price Similarity	.001	.000	.000	-.002	.009	.814	-.005	.003	.058	-.007	.003	.016	-.001	.002	.488
Deleted Brand Share	.005	.002	.012	.239	.049	.000	.030	.015	.042	-.160	.024	.000	.009	.009	.305
Competing Brand Share	-.002	.001	.001	.006	.017	.717	-.004	.004	.352	.013	.005	.013	.006	.004	.112
Category Concentration	.001	.001	.190	.010	.020	.619	-.006	.005	.206	.019	.007	.009	-.012	.006	.057
Category Growth	-.001	.009	.938	-.558	.397	.160	.008	.077	.918	.081	.102	.431	.100	.068	.138
Post x Moderators	✓			✓			✓			✓			✓		
Control Function Term	✓			✓			✓			✓			✓		
Cross-section FE	✓			✓			✓			✓			✓		
Time FE	✓			✓			✓			✓			✓		

Notes: Clustered standard errors by retailer-category-deleted brand-competing brand combination. SE = standard error. N = 55,876,309.

Relative to control markets, rival national brands (b=.022, p=.013) increase feature

advertising after a brand deletion. Private labels do so too (b=.048, p<.001), but only if there is no sister brand present (b=-.049, p=.013). Sister brands only increase their feature advertising if a private label is present (b=.058, p=.015). We find that competing brands especially increase feature advertising when the deleted brand's market share is higher (b=.239, p<.001).

We find all brand types to increase their regular price relative to control markets: +2.38% for rival national brands (b=.024, p<.001), +3.40% for sister brands (b=.033, p<.001), and +1.20% for private labels (b=.012, p<.001). Results for the moderators show that regular price changes are larger when the deleted brand had a higher market share (b=.030, p=.042), which is indicative of changes in the competitive structure as removing larger brands reduces competitive pressure more. Regular price changes are smaller when the competitor's price more closely matched the deleted brand's (b=-.005, p=.058). For private labels, regular price rises more if a sister brand is present (b=.009, p=.075); conversely, prices rise less for sister brands if private labels are present (b=-.007, p=.051).

On average, line length remains – compared to the control units – stable across all brand types. However, rival national brands' line length is reduced if private labels are present (b=-.007, p<.001), reflective of retailers trying to keep national brand power in check. Line length increases when the competing brand has a higher market share to start with (b=.015, p=.045) or the category is more concentrated (b=.019, p=.009); indicative of brands' greater power. However, line length declines when the deleted brand had a higher market share (b=-.160, p<.001), indicative of softened competition. Also, line length is reduced when the competing brand is priced more similarly (b=-.007, p=.016), possibly because a comparable brand continues to exist.

In response to the focal brand's deletion, all brand types increase their distribution: rival

national brand (b=.006, p<.001), sister brands (b=.011, p=.012), and private labels (b=.006, p<.001). However, private labels only do so if no sister brand is present (b=-.005, p=.079), and sister brands (b=-.009, p=.055) and rival national brands (b=-.002, p=.054) only do so if no private label is present. The moderator findings further show that lower distribution increases with higher category concentration (b=-.012, p=.057). Collectively, this hints at a hierarchy where retailers aim to first increase distribution of their private labels, followed by sister brands, which may have had more time to push for increased distribution compared to other competitors.

Thus far, we have assessed changes in marketing conduct at the average level of each continuous moderator. However, the brand types differ in the average values of the moderators, so we have to take the brand-specific values into account to calculate the net marketing conduct changes for each brand type. Thus, we next calculate the marketing conduct changes for all 63,037 market-retailer-category-brand cases by adding the intercept shifts for each brand type ($\hat{\beta}_{k,comp}$) and the impact of the moderators ($\sum_{q=1}^{M} \hat{\theta}_{kq}^{\Delta post,\Delta treat} MOD_{q,pbcmr}$), representing the sum of the observed moderator values multiplied by their respective coefficients. The marketing conduct change can be calculated as $\Delta\widehat{MKTG}_{k,bcmrt} \equiv \left(e^{(\hat{\beta}_{k,comp}+\sum_{q=1}^{M} \hat{\theta}_{kq}^{\Delta post,\Delta treat} MOD_{q,pbcmr})} - 1\right) * 100\%$. Table 8 shows the average percentage changes for each marketing variable by brand type. We observe the strongest average percentage lifts for regular price (especially for sister brands), followed by feature advertising and distribution.

Table 8. Net Marketing Conduct Changes by Brand Type

Brand Type	**Rival National Brand**	**Sister Brand**	**Private Label**	**Overall**
Cases	50,663	4,099	8,275	63,037
Promotion Price	-.05%	-.14%	-.01%	-.05%
Feature Advertising	1.92%	2.16%	4.21%	2.23%
Regular Price	2.44%	2.90%	1.37%	2.33%
Line Length	-.50%	.64%	-.38%	-.41%
Distribution	.42%	.59%	.49%	.44%

Net Revenue Effect

To establish the economic significance of marketing mix adjustments after a brand deletion, we calculate the additional revenue these adjustments generate compared to the baseline revenue change brands would experience without any marketing adjustments. To take the brand-type specific moderator values into account, we calculate the baseline effect, which is the brand-type specific gain due to the brand deletion ($\hat{\beta}_{comp}$ in Equation 1) plus the treatment effect of the moderators ($\sum_{q=1}^{M} \hat{\theta}_q^{\Delta post,\Delta treat} MOD_{q,pbcmr}$ in Equation 1), converted into a percentage change: $\left(e^{\hat{\beta}_{comp}+\sum_{q=1}^{M} \hat{\theta}_q^{\Delta post,\Delta treat} MOD_{q,pbcmr}} - 1\right) * 100\%$.

Table 9 shows the average baseline effect, separately for each competitor brand type (rival national brand, sister brand, and private label). We observe a .74% baseline revenue increase for rival national brands, a .51% increase for sister brands, and a 2.28% increase for private labels when not accounting for marketing conduct. Table 9 also shows the marketing conduct impact, which is the percentage revenue change due to marketing efforts after the brand deletion, calculated as the brand-type specific marketing elasticity ($\hat{\gamma}_{comp,k}$ from Equation 1) times the estimated marketing conduct change ($\Delta\widehat{MKTG}_{k,bcmrt}$) defined above for marketing variable k (1 = promotion price, 2 = feature advertising, 3 = regular price, 4 = line length, 5 = distribution).

Table 9. Net Revenue Changes by Competitor Brand Type

Brand Type	**Rival National Brand**	**Sister Brand**	**Private Label**	**Overall**
Cases	50,663	4,099	8,275	63,037
Baseline effect	.74%	.51%	2.28%	.93%
Promotion Price	.09%	.23%	.01%	.09%
Feature Advertising	.07%	.08%	.09%	.07%
Regular Price	-.12%	-.74%	-.25%	-.18%
Line length	-.63%	.84%	-.45%	-.51%
Distribution	1.23%	1.83%	1.77%	1.34%
Overall	**1.36%**	**2.77%**	**3.45%**	**1.73%**

Table 9 shows that the marketing conduct impact in the wake of a brand deletion is substantial. For sister brands, the observed increase in distribution adds 1.83% to revenue, while the line length adds .84%, which helps to boost the smaller baseline effect of .51%, leading to an overall revenue gain of 2.77% due to the brand deletion. Rival national brands manage to roughly double their baseline gain (+.74%) through marketing conduct (+.63%) to a total revenue gain of 1.36%, primarily through a distribution gain (+1.23%). Starting with a substantial baseline effect of +2.28%, some of the marketing conduct by private labels helps them (distribution increase adds 1.77%; feature advertising .09%), but other marketing conduct hurts: regular price hike reduces revenue by -.25%, line length reduction lowers revenue by -.45%. Still, private labels gain 3.45% in revenue in the wake of the brand deletion.

Considering the sizable marketing impact results in Table 9, we conclude that competing brands have considerable leverage to sway revenue gains their way through their marketing conduct after a brand deletion.

Net Revenue Implications for Brand Manufacturers

Using the net revenue effects for each competing brand (see Table 9), we assess how much sister brands gain after a brand deletion. For the manufacturer of both the sister brand and the deleted brand, the key question is whether the sister brand's gain covers a substantial share of the deleted brand's former revenue. In other words, can manufacturers succeed in "killing a brand, keeping a customer" (Kumar 2003, p. 86)? Although manufacturers may attempt to persuade customers to switch to sister brands in their brand portfolio, sister brands typically are positioned to minimize overlap and target distinct customer segments. Thus, customers are unlikely to view a sister brand as the most suitable alternative. Moreover, deleting a brand may create knowledge gaps for the manufacturer about how to serve customers.

To assess how a manufacturer's category revenues change when it deletes a brand, we calculate the cumulative revenue change across all remaining sister brands in the category for each market-retailer pair and compare it to the deleted brand's pretreatment revenue. The difference reveals the net revenue effect for the manufacturer. If the change is equivalent to the deleted brand's revenue, it retains 100%; if the revenue change for sister brands is smaller (larger), it retains less (more) than 100%. We then aggregate these results across all market-retailer pairs with at least one sister brand (on average, 1.36 sister brands were active).

For example, Wisk, a Sun Products brand that was deleted from the "Detergents - Heavy Duty – Liquid" category, earned $7.13M in annual national revenues in this category in our sample (conditional on a sister brand being present). After deletion, its Sun Products–owned sister brands gained $2.04M (on average, Wisk had with 3.79 sister brands), leading to a net revenue loss of -$5.09M (=$2.04M–$7.13M). Thus, the manufacturer recouped 28.52% (= [2.04/7.13]×100%) of the deleted brand's revenues through its sister brands. As Web Appendix M shows, across the 215 deleted brands with sister brands, manufacturers recoup on average 37.76% of deleted-brand revenue (the average deleted brand earned $485,799 per year per market-retailer pair, and sister brands recouped $183,453), though variation is substantial.

Results for Effect of Brand Deletion on Category Revenue

Table 10 reports on Equation 3, showing that the category revenue reduces by a net 2.3% (b=-.024, p<.001) in the wake of the focal brand's deletion. The model controls for the revenue effects of category-level regular price (b=.597, p<.001), feature advertising (b=.077, p<.001), and line length (b=.682, p<.001).

Table 10. Results for Effect of Brand Deletion on Category Revenue

Variable	Est.	SE	p-val.
Treat x Post	-.024	.006	.000
Post	.009	.003	.009
Category Price	.597	.074	.000
Category Feature	.077	.002	.000
Category Line Length	.682	.020	.000
Control Function Term	✓		
Cross-Section FE	✓		
Time FE	✓		

Notes: Clustered standard errors by retailer-category-deleted brand pairs. SE = standard error. N = 3,490,503

Additional Checks

We conduct several robustness checks to establish the sensitivity of our findings to modeling decisions. We describe these robustness checks in greater detail in Web Appendix N, where we also present detailed parameter estimates for the revenue model.

As a first robustness check, we assess the sensitivity of our revenue model results to the beginning of the brand deletion, which is determined using a moving average, by setting set T_0 (the beginning of the deletion) to two weeks before the week in our focal analyses and re-estimating our models. Second, we account for the staggered timing of treatment by employing a stacked DID approach. By carefully selecting control units that are never treated within the treated unit's timeframe, it avoids the bias that can arise in standard two-way fixed effects models, which mix valid and invalid comparisons (i.e., later-vs-earlier treated) when treatments are heterogeneous (Baker, Larcker, and Wang 2022; Goodman-Bacon 2021). As a robustness check, we apply a two-stage DID approach (Gardner et al. 2024) that accounts for staggered adoption. In the first stage, we remove group and time patterns using untreated observations to create adjusted outcomes; in the second stage, we estimate treatment effects on these adjusted values. This design eliminates problematic later-vs-earlier contrasts. Third, we construct counterfactuals for each treated brand using highly correlated control units. Our main analysis

requires an average correlation of $\bar{\rho} > .80$ revenue and marketing mix trajectories for the treated and control brands, ensuring close comparability. In two robustness checks, we use higher minimum average correlations of $\bar{\rho} > .85$ and $\bar{\rho} > .90$. Fourth, while our focal model uses clustered errors at the retailer-category-deleted brand-competing brand level, we also implement clustered standard errors at the panel (i.e., market-retailer-category-deleted brand-competing brand) level as an alternative to account for autocorrelation only.

Fifth, we assess to what extent our results in the revenue model are driven by the corrections for endogeneity. To this end, we estimate three additional versions of Equation 1: without any control-function terms, with only the control-function term for the deletion decision, and with only control-function terms for the marketing mix. Sixth, because brand deletions occur at different times across markets and retailers, we exploit this staggered timing by using not-yet-deleted cases as controls. To test whether timing matters, we compare early versus late deletions. A relevant question is whether effects differ between early versus late brand deletion cases. To investigate, we divide the sample into "Early deletion" cases (start date before the median) and "Late deletion" cases (after the median) and add an interaction effect of treatment x post with an indicator for Early deletion (base = Late deletion) in Equation 1. Seventh, it is conceivable that the five marketing mix variables have interactive effects and that there are decreasing returns to scale for line length. We added all 10 pairwise interactions to Equation 1, and separately added the squared log line length.

We find the parameter estimates to be very similar to the focal estimates, with an average correlation of .990 across robustness checks. We also note that the average proportion of estimates that are different in terms of significance (i.e., p-value just below vs. above .1) compared to the focal estimates (.06) is less than would be expected by chance (.1).

DISCUSSION

Brand manufacturers regularly prune their brand portfolios by deleting brands from a given product category. Despite the prevalence of brand deletions, their consequences are not well understood. We therefore adopt an empirics-first approach and assess – based on 1,046 brand deletions across 384 categories affecting 63k market-retailer-category-brand combinations – the effect of a brand's deletion on competing brands that remain active in the category. In so doing, we answer the four research questions introduced at the start.

First, we assess whether rival national brands, sister national brands, or private labels benefit from a brand's deletion. We find that – absent the impact of moderators and marketing conduct – rival national brands and sister brands experience no loss or gain, while private labels gain revenue in the wake of the brand deletion (Table 6; path A in Figure 1). When we add the impact of the moderators (research question 2; Table 9; path B in Figure 1), the revenue effect turns positive for all competing brands: rival national brands (+1.36%), sister brands (+2.77%), and private labels (+3.45%), for an average of +1.73%. The findings for the moderators add nuance to these insights, showing that competing brands gain more revenue when the competing brand had higher price similarity with the deleted brand, when the deleted brand's pre-deletion market share is higher, and when the category is less concentrated.

Further nuance is added by considering research question 3 about the extent to which competing brands change their marketing conduct after a brand deletion (path C in Figure 1). After accounting for the impact of moderators on marketing conduct (path D in Figure 1), we find that all brand types increase their regular price. However, because their price elasticity of revenue is negative, they experience a drop in revenue due to this move. All brand types manage to increase their distribution in the wake of the brand deletion, which, when combined with strong distribution elasticities, leads to reasonably sizeable revenue gains of 1.23%-1.83% due to

this form of marketing conduct across the three brand types. One surprising result is that only sister brands increase their line length in the wake of the brand deletion (Table 8), which (combined with a strong line length elasticity) generates an additional .84% revenue gain. A plausible explanation is that sister brands may have access to brand deletion information earlier, enabling them to plan line extensions in advance and thus react more quickly, preempting line extensions by others. In addition, they may benefit from reallocated resources, as the budget from the deleted brand can be used to support line length expansion.

Combining these insights allows us to address the final research question, which is how competing brands fare overall after the brand deletion, taking into account the extent to which they engage in marketing conduct changes (Table 9). We find that overall, private labels benefit the most (+3.45% revenue), with sister brands in second place (+2.77%) and rival national brands in third (+1.36%). Given that the vast majority of competing brands (80%) are rival national brands, the overall average revenue gain is 1.73%. We do note there is a lot of heterogeneity underlying these findings and a big role for marketing conduct, discussed next.

Marketing Conduct Insights

We document new insights with respect to brands' marketing conduct (relative to the control group) in the wake of a brand deletion, which we summarize in Table 11. We document four types of patterns. One pattern is a *lost opportunity* where marketing effectiveness is strong, but the marketing conduct change is minor, which we observe for price promotion for all brand types. A second pattern is *spoiled arms*, where marketing conduct change is in the right direction, but the effectiveness is weak, with feature advertising as a prime example, especially for private labels. A third pattern is a *seized opportunity*, where marketing conduct is adjusted in the right direction and the effectiveness is strong. The lift in distribution is a prime example for all three

brand types, as is the lift in line length for sister brands. The final and worst pattern is a *self-inflicted loss*, where an effective marketing variable is adjusted in the opposite direction from what would benefit revenue. The regular price hikes across all competitor brand types is one example of a self-inflicted loss, as are line length cuts by rival national brands and private labels.

Table 11. Insights on How Competing Brands Can Capitalize on Brand Deletions

	Marketing Mix	Effectiveness	Change in Conduct	Takeaway
Manufacturer-Driven Decisions	**Promotion Price**			
	Rival national brands	Strong price promotion elasticity (-1.765) makes promotion effective to increase revenue	Slightly deeper price promotion (-.01 % to -.14%)	*Lost opportunity* for all brand types to use more price promotion to lift revenue.
	Sister brands	Slightly less effective for sister brands (-1.685) and private labels (-1.591) than for rival national brands		
	Private labels			
	Feature Advertising			
	Rival national brands Sister brands	Weak revenue effect (~.035)	Moderate increase (~ 2%)	*Spoiled arms* (especially for private labels) by investing in this weak marketing instrument.
	Private labels	Even weaker tool than for rival and sister brands (.021)	Strong increase (+4.21%)	
Retailer Decisions with Manufacturers' Input	**Regular Price**			
	Rival national brands	Slightly negative elasticity of revenue to regular price (-.049)	Fairly strong regular price hike of 2.44%	A lift in regular price lowers revenue. All brand types do this to their disadvantage, causing *self-inflicted losses*.
	Sister brands	Most negative regular price elasticity (-.256)	Strongest regular price increase of 2.90%	
	Private labels	Moderately negative regular price elasticity (-.179)	Moderate regular price increase by 1.37%	
	Line Length			
	Rival national brands	Effective tool to lift revenue, with comparable elasticities of ~ 1.2 for all brand types	Small decrease (-.50%)	Increasing line length is a *seized opportunity* for sister brands, while rival national brands and private labels incur *self-inflicted losses* by cutting back on line length.
	Sister brands		Moderate increase (.64%)	
	Private labels		Small decrease (-.38%)	
Retailer Decisions	**Distribution**			
	Rival national brands Sister brands Private labels	Effective tool to lift revenue (elasticity ~ 3)	Small increase in distribution (.42% - .59%)	Distribution is a very effective tool to lift revenue, making distribution gains a *seized opportunity*.
		Very effective tool to lift revenue (3.586)		

Managerial Implications

Marketing literature is replete with advice on how to build brands (e.g., Ataman, Mela, and van Heerde 2008). In contrast, there is relatively little advice about how to deal with brand deletions. Doing nothing is a reasonable course of action for private labels (revenue +2.28%), but much less successful for rival national brands (+.74%) and particularly sister brands (+.51%). We recommend a more active response, and we highlight the promise of active marketing conduct, designed to capitalize on the opportunity created by a brand deletion (see also Table 9). One obvious caveat to the managerial implications presented next is that rival and sister brands may have different degrees of agency over marketing decisions. That is, while promotion price and feature advertising are primarily manufacturer-driven, regular price and line length are co-decided with retailers, while distribution is governed by retailers. We also note that our recommendations focus on revenue (rather than profit) implications of marketing conduct, as we lack data on the cost of changes in marketing and on profit margins.

Rival national brands

For rival national brands, deploying price promotion is a feasible, effective, yet underutilized form of marketing conduct in the wake of a brand deletion (see Table 11). Even if price promotions might threaten to undermine a premium brand image or brand equity in the long term (Jedidi, Mela, and Gupta 1999), they can strongly boost brand revenue in the short term to seize customers from the deleted brand. Feature advertising, in contrast, does not necessarily need to be increased given its relatively weak effectiveness.

When rival national brands decide with retailers on regular price in the wake of the brand deletion, one important input into this decision is the relatively inelastic price response (-.049) of the revenue of rival national brands. In a way, the inelastic response justifies the fairly strong

regular price rise we document for these brands (+2.44%). As for line length, rival national brands should point out to retailers that the strong line length elasticities (around 1.3) warrant extending the brand's line length carried by the retailer, which is opposite to the shrinkage (-.50%) that we document empirically. To convince retailers to carry the brand in more stores, rival national brand manufacturers should point out the strong distribution elasticity (2.942).

Sister brands

The recommendations for sister brands are largely parallel to those for rival national brands. They should also offer more price promotions (without a need for feature advertising) and point to strong line length and distribution elasticities to convince retailers to carry their SKUs in more stores. One diverging implication emerges for regular price: given the relatively strong elasticity of revenue to regular price for sister brands (-.256), the observed strong increase in regular price (+2.90%) caused self-inflicted revenue losses. Mitigating regular price increases (or even reducing regular price after a brand deletion) is recommended for sister brands.

Looking at the bigger picture, for manufacturers that maintain multiple brands in a category and plan to delete one, our findings provide cold comfort. Sister brands recoup less than 40% of the revenue lost due to the brand deletion on average. Of course, other considerations (unobservable to us) might prompt manufacturers to delete a brand from a category (e.g., efficiency gains in brand management, supply issues), but from an overall revenue perspective, we suggest caution.

Private labels

Our findings are especially encouraging for private labels. Private labels gain, on average, 2.28% revenue from a brand deletion if they keep their marketing mix unchanged. Moreover, private labels have full control over all marketing variables as these brands are owned

by retailers. Capitalizing on their strong distribution elasticity (3.586), private labels should increase their distribution (if it is not at 100% yet), seizing the shelf space left vacant by the deleted brand. Private labels' second and third strongest elasticities are for price promotion (-1.591) and line length (1.190), making price promotion and line length extension effective (yet currently underutilized) ways to attract customers in the wake of the brand deletion. Given feature advertising's relatively weak elasticity (.021), we again suggest caution. Although feature advertising is relatively easy to deploy as a marketing tactic (and private labels eagerly do so), we find it is not an effective way to gain revenue after brand deletion.

Overall, while private label marketing effectiveness is somewhat weaker on average than sister brands and rival national brands (except for distribution, where they dominate national brands), they can gain a lot of ground by making judicious adjustments to their most powerful marketing variables, such as distribution, price promotion, and line length. Thus, while an inactive stance in response to a brand deletion will bring some benefits to private labels, an active stance (i.e., increasing marketing support to capitalize on the deletion) can potentially produce strong payoffs.

Limitations and Further Research

While we shed some first light on the consequences of brand deletions, we suggest that follow-up research could investigate their antecedents (which is beyond the scope of this study). A variety of possible motives might drive manufacturers to delete brands from the market, including economic (lack of profitability), branding (creating opportunities for sister brand growth), supply (difficulty of sourcing inputs), knowledge (lack of in-firm category expertise), or brand consistency (deleted brand does not fit the rest of the brand portfolio) motives. Investigating the prevalence and strength of these different motives for brand deletions can shed

additional light on this under-researched domain. The decision to delete a brand has potentially many antecedents, some of which are observable (e.g., the presence of sister brands) but others are unobservable (change in focus, anticipation of competitors).

Unfortunately, we lack advertising data for all markets, retailers, categories, brands, and time periods. However, recent research studying U.S. grocery products reports very small advertising elasticities (.0098-.0261; Shapiro, Hitsch, and Tuchman 2021), suggesting minimal advertising effects. We also note that national advertising influences both treated and control units, so even with such data, isolating a treatment effect would not be feasible.

Furthermore, we document the revenue consequences of brand deletions using retail scanner data, highlighting a promising avenue for examining how brand pruning affects investor and firm value. In the introduction, we reference large-scale pruning initiatives by firms such as P&G (100 brands deleted) and Coca-Cola (200 brands). Future research could investigate how these strategies influence firm value (e.g., abnormal returns or Tobin's Q). Such research could quantify both the magnitude and mechanisms of these effects – for instance, whether cost savings offset potential revenue losses.

Our sample frame covers each market's top 10 competing brands and each market's top 3 retailers to ensure enough economic substance. Given this decision rule, we had to omit smaller competing brands that are not among the 10 leading competing brands in the market, but may be a top 10 competing brand within a given retailer.

In summary, we believe this study makes some meaningful headway in understanding the consequences of brand deletions and how competing brands can seize the opportunities by engaging with their marketing mix. We further hope this study stimulates new research into this important yet underexplored marketing domain.

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Web Appendix

Filling the Void: How Competing Brands Can Capitalize on a Brand Deletion

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Web Appendix A: List of Categories

Group Description	Module Description (Category)
Baby Food	Baby Food - Strained
Baby Food	Baby Food - Junior
Baby Needs	Baby Accessory
Baby Needs	Nursing Accessories
Baby Needs	Baby Care Products-Bath
Baked Goods-Frozen	Bakery - Biscuits/Rolls/Muffins - Frozen
Baked Goods-Frozen	Bakery-Bread-Frozen
Baked Goods-Frozen	Bakery - Dessert Cakes - Frozen
Baking Mixes	Mixes - Brownies
Baking Mixes	Mixes-Dessert-Misc.
Baking Mixes	Mixes-Rolls & Biscuits
Baking Mixes	Mixes - Pancake
Baking Supplies	Croutons
Baking Supplies	Cake Decorations & Icing
Baking Supplies	Coconut
Baking Supplies	Cocoa
Batteries And Flashlights	Batteries
Beer	Beer
Beer	Light Beer (Low Calorie/Alcohol)
Beer	Ale
Books And Magazines	Magazines Selected Titles
Bread And Baked Goods	Bakery - Bread - Fresh
Bread And Baked Goods	Bakery-Buns-Fresh
Bread And Baked Goods	Bakery-Rolls-Fresh
Bread And Baked Goods	Bakery-Muffins-Fresh
Bread And Baked Goods	Bakery-Cakes-Fresh
Bread And Baked Goods	Bakery-Breakfast Cakes/Sweet Rolls-Fresh
Bread And Baked Goods	Bakery-Doughnuts-Fresh
Bread And Baked Goods	Bakery-Pies-Fresh
Bread And Baked Goods	Bakery-Bagels-Fresh
Bread And Baked Goods	Bakery - Remaining - Fresh
Bread And Baked Goods	Bakery-Cheesecake-Fresh
Breakfast Food	Toaster Pastries
Breakfast Food	Granola & Yogurt Bars
Breakfast Food	Breakfast Bars
Breakfast Foods-Frozen	Frozen/Refrigerated Breakfasts
Breakfast Foods-Frozen	Frozen Waffles & Pancakes & French Toast
Butter And Margarine	Margarine And Spreads
Candy	Breath Sweeteners
Candy	Candy-Chocolate-Miniatures
Candy	Candy-Chocolate
Candy	Candy-Chocolate-Special
Candy	Candy-Non-Chocolate
Candy	Candy-Lollipops
Candy	Candy-Dietetic - Non-Chocolate
Candy	Candy-Dietetic - Chocolate
Carbonated Beverages	Soft Drinks - Carbonated
Carbonated Beverages	Soft Drinks - Low Calorie
Cereal	Cereal - Ready To Eat
Cereal	Cereal - Granola & Natural Types
Cereal	Cereal - Hot
Charcoal, Logs, Accessories	Charcoal
Cheese	Cheese - Natural - Mozzarella
Cheese	Cheese - Natural - Remaining
Cheese	Cheese - Natural - American Cheddar
Cheese	Cheese - Grated
Cheese	Cheese - Processed - Snack
Cheese	Cheese-Processed Slices-American
Cheese	Cheese-Natural-Swiss
Cheese	Cheese - Specialty/Imported
Cheese	Cheese - Natural - Variety Pack
Cheese	Cheese - Shredded
Cheese	Cheese - Processed - Cream Cheese
Coffee	Ground And Whole Bean Coffee
Coffee	Coffee - Soluble Flavored
Coffee	Coffee - Liquid
Condiments, Gravies, And Sauces	Barbecue Sauces
Condiments, Gravies, And Sauces	Spaghetti/Marinara Sauce
Condiments, Gravies, And Sauces	Cooking Sauce
Condiments, Gravies, And Sauces	Sauces - Miscellaneous - Shelf Stable
Condiments, Gravies, And Sauces	Gravy - Canned
Condiments, Gravies, And Sauces	Sauces - Dipping
Condiments, Gravies, And Sauces	Sauce Mix - Taco
Condiments, Gravies, And Sauces	Mustard
Cookies	Cookies
Cookware	Cookware Product
Cosmetics	Cosmetics-Lipsticks
Cosmetics	Cosmetics-Eyebrow & Eye Liner
Cosmetics	Cosmetics-Nail Polish
Cosmetics	Cosmetics-Mascara
Cosmetics	Cosmetic Kits
Cosmetics	Cosmetics-Remaining
Cosmetics	Cosmetics-Foundation-Cream And Powder
Cosmetics	Cosmetics - Concealers
Cot Cheese, Sour Cream, Toppings	Dairy - Sour Cream - Refrigerated & Canned
Cot Cheese, Sour Cream, Toppings	Cheese - Cottage
Cough And Cold Remedies	Nasal Product Internal
Cough And Cold Remedies	Cold Remedies - Adult
Cough And Cold Remedies	Cough Syrups & Tablets
Cough And Cold Remedies	Sinus Remedies
Crackers	Crackers - Cheese
Crackers	Wafers & Toast & Bread Sticks
Crackers	Crackers - Flavored Snack
Deodorant	Deodorants - Personal
Deodorant	Deodorants - Cologne Type
Desserts, Gelatins, Syrup	Syrup-Specialty
Desserts, Gelatins, Syrup	Toppings-Liquid & Dry
Desserts, Gelatins, Syrup	Desserts - Rts Single Servings - Canned
Desserts/Fruits/Toppings-Frozen	Frozen Fruits
Desserts/Fruits/Toppings-Frozen	Frozen Desserts
Detergents	Detergents-Packaged
Detergents	Detergents - Light Duty
Detergents	Detergents - Heavy Duty - Liquid
Detergents	Automatic Dishwasher Compounds
Diet Aids	Dieting Aids-Complete Nutritional
Disposable Diapers	Disposable Diapers
Dressings/Salads/Prep Foods-Deli	Pasta - Refrigerated
Dressings/Salads/Prep Foods-Deli	Precut Fresh Salad Mix
Dressings/Salads/Prep Foods-Deli	Fruit-Refrigerated
Dressings/Salads/Prep Foods-Deli	Fruit Salads-Refrigerated
Dressings/Salads/Prep Foods-Deli	Remaining-Ready Made Salads
Dressings/Salads/Prep Foods-Deli	Meat & Sandwich Spreads-Refrigerated
Dressings/Salads/Prep Foods-Deli	Entrees-Refrigerated
Dressings/Salads/Prep Foods-Deli	Sandwiches-Refrigerated/Frozen
Dressings/Salads/Prep Foods-Deli	Combination Lunches
Dressings/Salads/Prep Foods-Deli	Pizza-Refrigerated
Eggs	Eggs-Fresh
Electronics, Records, Tapes	Ink Jet And Toner Cartridge
Electronics, Records, Tapes	Disk Diskette And Data Cartridge
Electronics, Records, Tapes	Printers
Ethnic Haba	Hair Preparations-Ethnic
Feminine Hygiene	Feminine Hygiene-Miscellaneous
First Aid	First Aid - Treatments
First Aid	Adhesive Bandages - Liquid - Powder - Paste
First Aid	First Aid - Ice And Heat Pack
Floral, Gardening	Lawn And Soil Fertilizer And Treatment
Fragrances - Women	Cologne & Perfume-Women's
Fragrances - Women	Children's Cologne & Gift Sets
Fresh Meat	Fresh Meat
Fresh Produce	Fresh Apples
Fresh Produce	Fresh Herbs
Fresh Produce	Fresh Mushrooms
Fresh Produce	Fresh Carrots
Fresh Produce	Fresh Celery
Fresh Produce	Fresh Fruit-Remaining
Fresh Produce	Fresh Vegetables-Remaining
Fresh Produce	Fresh Onions
Fresh Produce	Fresh Potatoes
Fresh Produce	Fresh Strawberries
Fresh Produce	Fresh Tomatoes
Fresheners And Deodorizers	Air/Specialty Freshners - Aerosol Spray & Pump
Fresheners And Deodorizers	Air/Specialty Fresheners - Remaining
Fruit - Dried	Fruit-Dried And Snacks
Glassware, Tableware	Candle And Candle In Holder
Glassware, Tableware	Drinkware Container Set
Grooming Aids	Hair Care And Fashion Accessory
Grooming Aids	Cosmetics - Applicator Brushes
Grt Cards/Party Needs/Novelties	Candle Holder And Accessory
Gum	Gum-Bubble
Gum	Gum-Chewing-Sugarfree
Hair Care	Shampoo-Aerosol/ Liquid/ Lotion/ Powder
Hair Care	Creme Rinses & Conditioners
Hair Care	Wave Setting Products
Hair Care	Hair Spray - Women's
Hair Care	Hair Coloring - Women's
Hair Care	Hair Preparations - Other Than Men's
Hardware, Tools	Storage And Space Management
Hardware, Tools	Trash And Recycling Bin
Hardware, Tools	Buckets & Pails
Household Cleaners	Cleaners - Disinfectants
Household Cleaners	Cleaners - Non-Disinfectant
Household Cleaners	Cleaners - Bathroom
Household Supplies	Bathroom Accessory
Household Supplies	Scouring Pads
Household Supplies	Brushes-Bathroom
Household Supplies	Thermometers-Household/Outdoor
Household Supplies	Water Filtration Storage Container
Household Supplies	Cloth-Polishing/Cleaning
Household Supplies	Brooms/ Mops & Wax Applicators
Household Supplies	Dustpans
Housewares, Appliances	Household Appliance Accessories
Housewares, Appliances	Garment Steamer And Iron Appliance
Housewares, Appliances	Coffee And Tea Maker Appliance
Housewares, Appliances	Blender Appliance
Housewares, Appliances	Juicer Appliance
Housewares, Appliances	Cooker Steamer And Dehydrator Appliance
Housewares, Appliances	Microwave Oven Appliance
Housewares, Appliances	Refrigerator And Freezer Appliance
Housewares, Appliances	Vacuum And Carpet Cleaner Appliance
Housewares, Appliances	Humidifier And Vaporizer Appliance
Housewares, Appliances	Fan And Ceiling Fan Appliance
Housewares, Appliances	Household Specialty Appliance
Housewares, Appliances	Hair Styling Appliance And Accessory
Housewares, Appliances	Oral Hygiene Appliance And Accessory
Housewares, Appliances	Razor Trimmer Appliance And Accessory
Housewares, Appliances	Skin And Nail Appliance And Accessory
Ice Cream, Novelties	Yogurt - Frozen
Ice Cream, Novelties	Ice Cream - Bulk
Ice Cream, Novelties	Frozen Novelties
Insecticds/Pesticds/Rodenticds	Insecticide - Remaining Miscellaneous Products
Jams, Jellies, Spreads	Jams
Jams, Jellies, Spreads	Butter-Fruit & Honey
Jams, Jellies, Spreads	Honey
Jams, Jellies, Spreads	Peanut Butter
Juice, Drinks - Canned, Bottled	Fruit Drinks & Juices-Cranberry
Juice, Drinks - Canned, Bottled	Fruit Juice - Lemon/Lime
Juice, Drinks - Canned, Bottled	Fruit Juice - Orange - Other Container
Juice, Drinks - Canned, Bottled	Fruit Drinks-Canned
Juice, Drinks - Canned, Bottled	Fruit Drinks-Other Container
Juice, Drinks - Canned, Bottled	Fruit Juice-Remaining
Juice, Drinks - Canned, Bottled	Vegetable Juice And Drink Remaining
Juices, Drinks-Frozen	Fruit Drinks & Mixes - Frozen
Kitchen Gadgets	Dishwashing Accessory Product
Kitchen Gadgets	Kitchen Accessory Product
Kitchen Gadgets	Kitchen Utensil And Gadget
Kitchen Gadgets	Food Storage Containers
Kitchen Gadgets	Beverage Storage Container
Kitchen Gadgets	Kitchen Cutlery And Flatware
Laundry Supplies	Fabric Softeners-Dry
Laundry Supplies	Laundry & Ironing Accessories
Light Bulbs, Electric Goods	Telephone And Accessory
Light Bulbs, Electric Goods	Lamps - Incandescent
Liquor	Canadian Whiskey
Liquor	Vodka
Liquor	Rum
Liquor	Tequila
Liquor	Alcoholic Cocktails
Liquor	Coolers - Remaining
Medications/Remedies/Health Aids	Pregnancy Test Kits
Medications/Remedies/Health Aids	Tooth & Gum Analgesics
Medications/Remedies/Health Aids	Antacids
Medications/Remedies/Health Aids	Pain Remedies - Headache
Medications/Remedies/Health Aids	Laxatives
Medications/Remedies/Health Aids	Lip Remedies - Solid

Group Description	Module Description (Category)
Medications/Remedies/Health Aids	Sleeping Aids
Medications/Remedies/Health Aids	Insoles
Medications/Remedies/Health Aids	Foot Preparations-Remaining
Medications/Remedies/Health Aids	Foot Comforts Products
Medications/Remedies/Health Aids	Pain Remedies - Children's Liquid
Medications/Remedies/Health Aids	Adult-Incontinence
Medications/Remedies/Health Aids	Medical Accessory-Remaining
Milk	Dairy-Flavored Milk-Refrigerated
Milk	Dairy-Milk-Refrigerated
Milk	Dairy-Cream-Refrigerated
Milk	Remaining Drinks & Shakes-Refrigerated
Nuts	Nuts - Cans
Nuts	Nuts - Jars
Nuts	Nuts - Bags
Nuts	Nuts - Unshelled
Oral Hygiene	Oral Hygiene Brushes
Oral Hygiene	Tooth Cleaners
Oral Hygiene	Denture Cleansers
Oral Hygiene	Tooth Whiteners
Packaged Meats-Deli	Sausage-Dinner
Packaged Meats-Deli	Lunchmeat-Sliced-Refrigerated
Packaged Meats-Deli	Frankfurters-Refrigerated
Packaged Meats-Deli	Bacon-Refrigerated
Packaged Meats-Deli	Sausage-Breakfast
Packaged Meats-Deli	Lunchmeat-Deli Pouches-Refrigerated
Packaged Milk And Modifiers	Milk-Shelf Stable
Paper Products	Paper Napkins
Paper Products	Toilet Tissue
Paper Products	Disposable Cups
Paper Products	Disposable Dishes
Paper Products	Paper Towels
Pasta	Oriental Noodles
Pasta	Pasta - Noodles & Dumplings
Personal Soap And Bath Additives	Soap - Liquid
Personal Soap And Bath Additives	Soap - Specialty
Personal Soap And Bath Additives	Hand Cleaners And Hand Sanitizers
Pet Care	Pet Accessory
Pet Care	Pet Treatment Internal
Pet Care	Cat & Dog Litter
Pet Care	Pet Care - Rawhide & Chew Products
Pet Food	Pet Care - Pet Food
Pet Food	Cat Food - Dry Type
Pet Food	Dog Food - Wet Type
Pet Food	Dog Food - Dry Type
Pet Food	Dog & Cat Treats
Photographic Supplies	Cameras
Pickles, Olives, And Relish	Olives - Green
Pickles, Olives, And Relish	Pickles - Dill
Pizza/Snacks/Hors Doeurves-Frzn	Frozen/Refrigerated Hors D' Oeuvres & Snacks
Pizza/Snacks/Hors Doeurves-Frzn	Pizza-Frozen
Prepared Food-Dry Mixes	Mexican Dinners-Dry/Kit
Prepared Food-Dry Mixes	Mexican Tortillas
Prepared Food-Dry Mixes	Pizza Pie And Crust Mixes
Prepared Food-Dry Mixes	Rice - Mixes
Prepared Food-Dry Mixes	Dry Dinners - Pasta
Prepared Food-Dry Mixes	Mixes - Ethnic Specialties
Prepared Food-Ready-To-Serve	Entrees/Side Dishes - Shelf Stable
Prepared Food-Ready-To-Serve	Pickled Vegetables & Fruit
Prepared Food-Ready-To-Serve	Spreads-Hors D'oeuvres
Prepared Food-Ready-To-Serve	Spaghetti - Canned
Prepared Food-Ready-To-Serve	Oriental Foods-Misc.
Prepared Foods-Frozen	Meal Starters
Prepared Foods-Frozen	Entrees - Poultry - 2 Food - Frozen
Prepared Foods-Frozen	Dinners-Frozen
Prepared Foods-Frozen	Pot Pies-Frozen
Prepared Foods-Frozen	Entrees - Meat - 1 Food - Frozen
Prepared Foods-Frozen	Entrees - Poultry - 1 Food - Frozen
Prepared Foods-Frozen	Entrees - Oriental - 1 Food - Frozen
Prepared Foods-Frozen	Entrees - Italian - 1 Food - Frozen
Prepared Foods-Frozen	Entrees - Mexican - 1 Food - Frozen
Prepared Foods-Frozen	Entrees - Remaining - 1 Food - Frozen
Prepared Foods-Frozen	Entrees - Seafood - 1 Food - Frozen
Prepared Foods-Frozen	Pasta-Plain-Frozen
Prepared Foods-Frozen	Soup-Frozen-Refrigerated
Prepared Foods-Frozen	Sauces & Gravies-Frozen/Refrigerated
Pudding, Desserts-Dairy	Pudding-Refrigerated
Salad Dressings, Mayo, Toppings	Salad & Potato Toppings-Dry
Salad Dressings, Mayo, Toppings	Salad Dressing - Liquid
Sanitary Protection	Tampons
Seafood - Canned	Seafood - Sardines - Canned
Shaving Needs	Depilatories - Women's
Shaving Needs	Razors Disposable
Shortening, Oil	Salad And Cooking Oil
Shortening, Oil	Olive Oil
Skin Care Preparations	Suntan Preparations - Lotions/ Oils/ Etc.
Skin Care Preparations	Face Cleansers & Creams & Lotions
Skin Care Preparations	Hand & Body Lotions
Skin Care Preparations	Skin Bleaching/Toning Products
Skin Care Preparations	Skin Cream-All Purpose
Skin Care Preparations	Acne Remedies
Snacks	Dip - Canned
Snacks	Snacks - Meat
Snacks	Snacks - Puffed Cheese
Snacks	Snacks - Potato Chips
Snacks	Snacks - Corn Chips
Snacks	Snacks - Tortilla Chips
Snacks	Snacks - Remaining
Snacks	Popcorn - Popped
Snacks	Popcorn - Unpopped
Snacks	Snacks - Pretzel
Snacks	Snacks - Caramel Corn
Snacks	Snacks - Variety Packs
Snacks	Crackers - Sandwich & Snack Packs
Snacks	Trail Mixes
Snacks	Snacks - Health Bars & Sticks
Snacks, Spreads, Dips-Dairy	Seafood-Refrigerated
Snacks, Spreads, Dips-Dairy	Spreads-Remaining
Snacks, Spreads, Dips-Dairy	Dairy - Dip - Refrigerated & Frozen
Soft Drinks-Non-Carbonated	Remaining Drinks & Shakes-Non Refrigerated
Soft Drinks-Non-Carbonated	Soft Drinks - Powdered
Soft Drinks-Non-Carbonated	Ice Pops - Unfrozen
Soft Drinks-Non-Carbonated	Cocktail Mixes-Liquid
Soft Drinks-Non-Carbonated	Water-Bottled
Soup	Soup-Canned
Soup	Soup Mixes - Dry & Bases
Stationery, School Supplies	Artist And Hobby Paint And Supply
Stationery, School Supplies	Tape - Miscellaneous
Stationery, School Supplies	School And Office Paper And Forms
Stationery, School Supplies	Correction Fluid And Erasers
Stationery, School Supplies	Personal Planners Binders And Folders
Stationery, School Supplies	School And Office Storage And Dispenser
Stationery, School Supplies	Home School And Office Combinations
Stationery, School Supplies	Adhesive Note Pads
Stationery, School Supplies	Pencils - Colored
Stationery, School Supplies	Markers
Stationery, School Supplies	Pens & Pencils
Stationery, School Supplies	Paper Shredders
Sugar, Sweeteners	Sugar Substitutes
Table Syrups, Molasses	Syrup - Table
Tea	Tea - Instant
Tea	Tea - Liquid
Tobacco & Accessories	Lighters
Tobacco & Accessories	Cigarettes
Tobacco & Accessories	Tobacco-Smoking
Tobacco & Accessories	Anti-Smoking Products
Unprep Meat/Poultry/Seafood-Frzn	Frozen Poultry
Unprep Meat/Poultry/Seafood-Frzn	Seafood - Shrimp - Unbreaded - Frozen
Unprep Meat/Poultry/Seafood-Frzn	Seafood-Fish-Unbreaded-Frozen
Unprep Meat/Poultry/Seafood-Frzn	Frozen Meat - Ground Beef
Vegetables - Canned	Vegetables-Artichokes-Canned
Vegetables - Canned	Vegetables-Corn-Whole Kernel-Canned
Vegetables - Canned	Vegetables-Onions-Canned
Vegetables - Canned	Tomatoes - Remaining - Canned
Vegetables And Grains - Dried	Rice - Packaged And Bulk
Vegetables-Frozen	Vegetables - Corn - Frozen
Vegetables-Frozen	Vegetables - Potatoes - Frozen/Refrigerated
Vegetables-Frozen	Vegetables-In Sauce-Frozen
Vegetables-Frozen	Vegetables - Mixed - Frozen
Vitamins	Nutritional Supplements
Vitamins	Complete Nutritional Products
Vitamins	Vitamins-Children-Flavored Chewable
Vitamins	Vitamins-Remaining
Vitamins	Protein Supplements
Wine	Wine-Domestic Dry Table
Wine	Wine-Imported Dry Table
Wine	Wine-Flavored/Refreshment
Wrapping Materials And Bags	Bags - Trash/Trash Compactor
Wrapping Materials And Bags	Bags - Tall Kitchen
Yogurt	Yogurt-Refrigerated
Yogurt	Yogurt-Refrigerated-Shakes & Drinks

Web Appendix B: Summary Statistics

Figure W1. Histogram of Length of Brand Deletions (N = 14,462 Market-Retailer-Category-Deleted Brand Combinations)

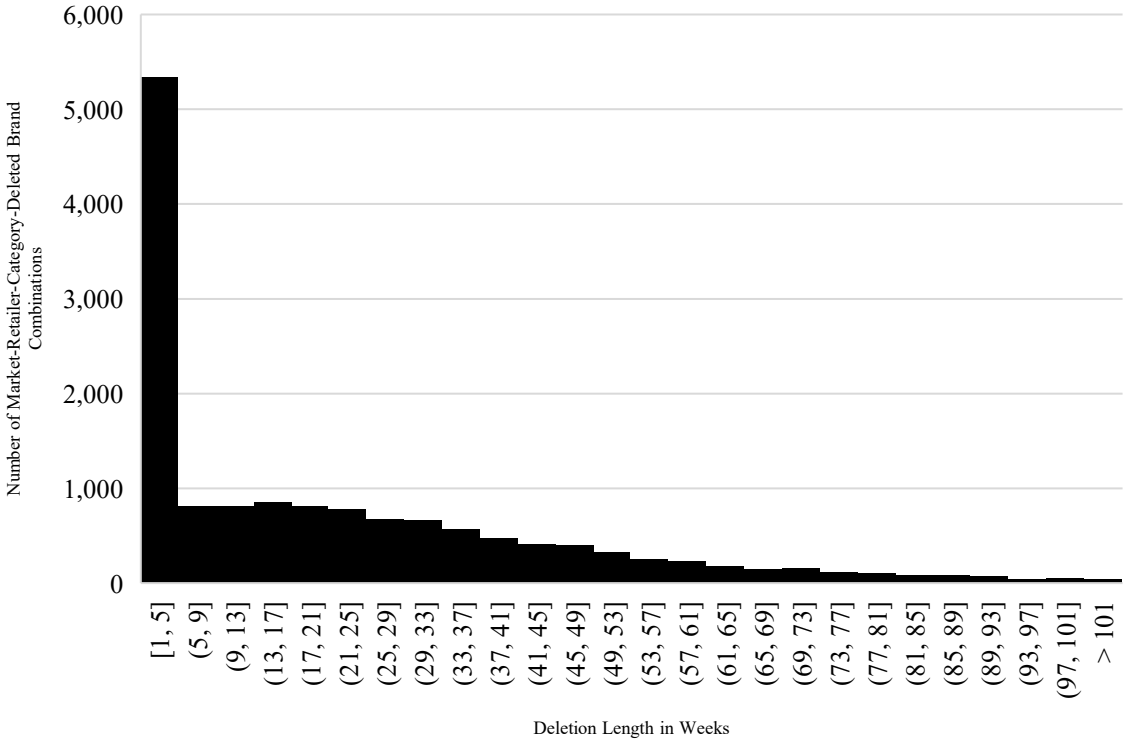


Table W1. Number of Observations, Means, and Standard Deviations by Competitor Brand Type and Marketing Mix

	# Observations (at the Market-Retailer-Category-Brand-Week Level)	Weekly Revenue	Promotion Price (Ratio of Actual to Regular Price x 100)	Feature Advertising (Weighted Percentage across SKUs and Stores)	Regular Price ($)	Line Length	Distribution in %
Rival National Brands	44,753,732	3,208.05	95.59	6.06	8.47	11.41	98.71
		(9,760.00)	(6.88)	(20.13)	(62.51)	(13.46)	(6.75)
Sister Brands	3,767,662	3,871.62	94.69	7.26	3.12	10.49	99.13
		(8,993.71)	(7.85)	(21.45)	(16.55)	(9.55)	(5.66)
Private Label	7,354,915	5,764.38	96.12	5.75	2.06	16.61	99.57
		(18,945.95)	(4.86)	(17.28)	(6.18)	(17.13)	(3.94)
Overall	55,876,309	3,589.28	95.60	6.10	7.27	12.03	98.85
		(11,390.45)	(6.73)	(19.88)	(56.20)	(13.90)	(6.39)

Notes: Values represent averages, values in parentheses are standard deviations. Regular Price is not comparable across categories. To provide some insights into the relative price levels for the three brand types, we calculate for each of the 14,462 market-retailer-category-deleted brand combinations the regular price ratio of sister brands and private labels to the average rival national brand. On average, sister brands (private labels) have a 11% higher (14% lower) regular price than rival national brands.

Web Appendix C: Correlation Matrix of Key Variables in Brand Revenue Model (Equation 1)

	Variable	1	2	3	4	5	6	7	8	9	10	11	12	13
1	**Revenue**	1												
2	**Promotion Price**	-.101	1											
3	**Feature Advertising**	.232	-.265	1										
4	**Regular Price**	.008	.069	-.030	1									
5	**Line Length**	.444	-.023	.143	.151	1								
6	**Distribution**	.226	-.017	.031	-.033	.128	1							
7	**Sister Brand Present**	.091	-.018	.036	-.147	.026	.018	1						
8	**Private Label Present**	.142	-.007	.029	-.111	.042	.013	.057	1					
9	**Price Similarity**	.070	-.021	.028	-.091	.066	.022	.102	.039	1				
10	**Deleted Brand Share**	-.143	.052	-.052	.042	-.189	-.020	-.117	-.128	-.174	1			
11	**Competing Brand Share**	.259	-.003	.103	.055	.349	.076	-.112	-.043	-.184	.249	1		
12	**Category Concentration**	-.028	-.011	-.010	-.023	-.066	-.035	-.135	.015	-.275	.226	.528	1	
13	**Category Growth**	.001	.009	-.018	-.033	.002	.009	.020	.028	.015	-.072	-.023	-.023	1

Notes: All correlations are significant at $p < .001$.
N = 55,876,309.

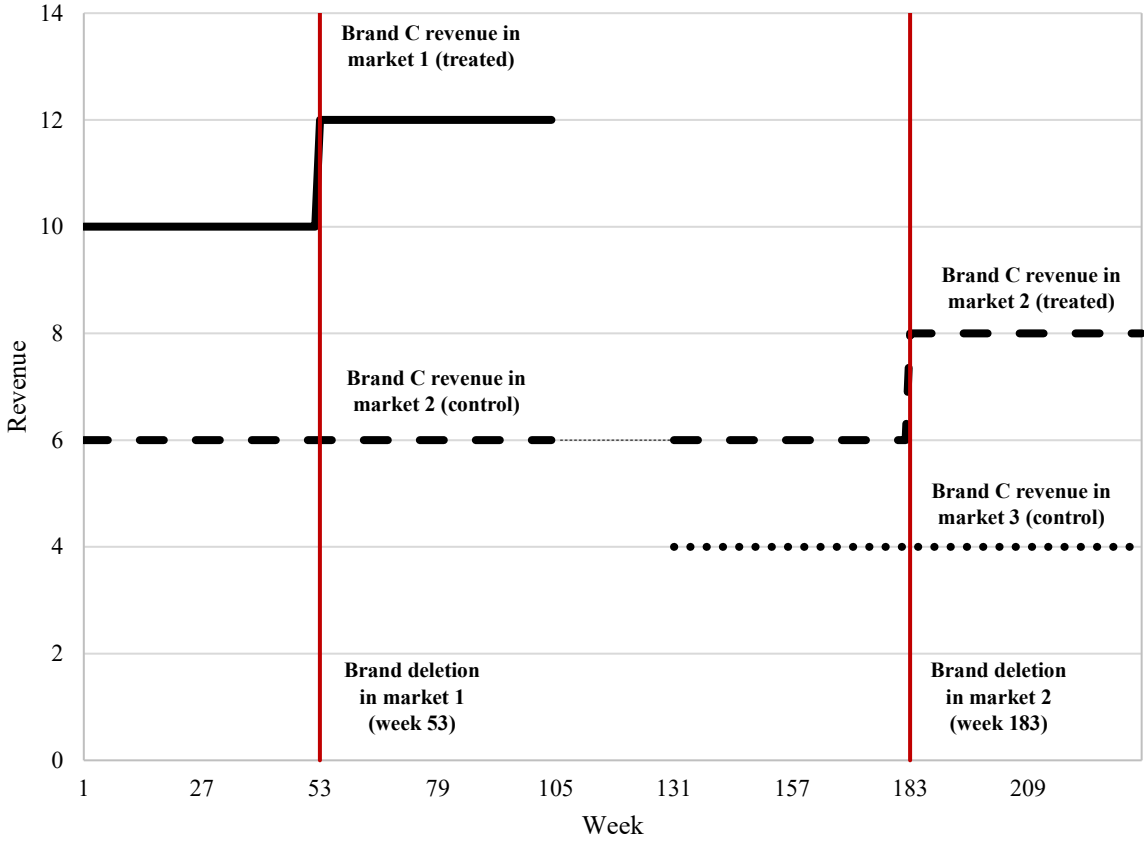
Web Appendix D: Illustrative Data Structure of Treated and Control Cases

In this illustrative example, we consider the deletion of brand D by retailer R in markets M1 and M2. Brand D is deleted in market M1 (M2) in week 53 (183). Market M2 serves as control unit for market M1; market M3 (to be deleted even later) serves as control unit for market M2. The data structure shows that while M2 serves as control unit (weeks 1 – 105), it is "never treated". Table W2 and Figure W2 illustrate the data structure.

Table W2. Illustrative Data Structure

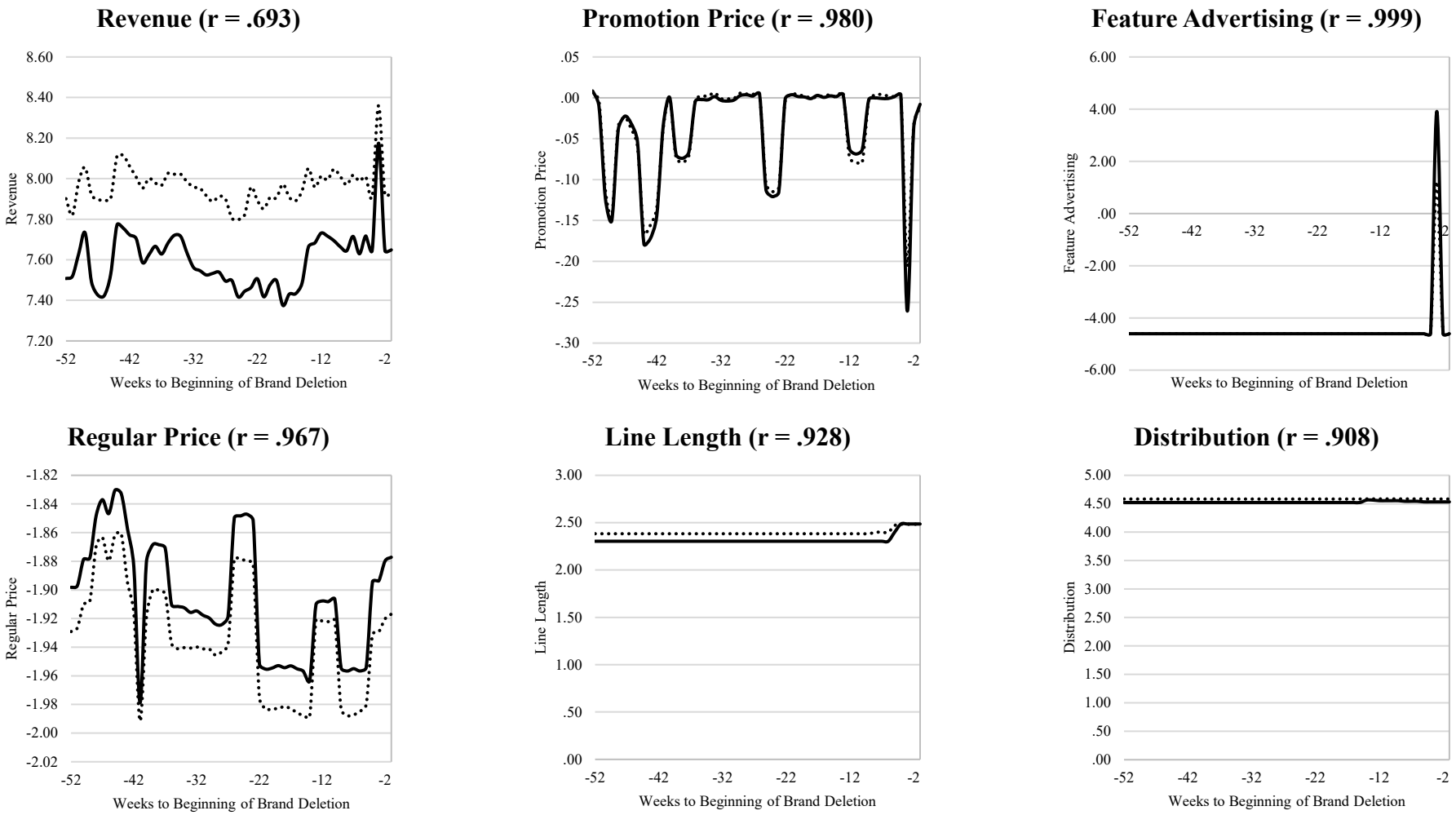
"Case"	Deleted Brand	Competing Brand	Retailer	Market	Time	Treat	Post
1	D	C	R	M1	1	1	0
1	D	C	R	M1	2	1	0
1	D	C	R	M1	…		
1	D	C	R	M1	52	1	0
1	D	C	R	M1	53	1	1
1	D	C	R	M1	54	1	1
1	D	C	R	M1	…		
1	D	C	R	M1	105	1	1
1	D	C	R	M2	1	0	0
1	D	C	R	M2	2	0	0
1	D	C	R	M2	…		
1	D	C	R	M2	52	0	0
1	D	C	R	M2	53	0	1
1	D	C	R	M2	54	0	1
1	D	C	R	M2	…		
1	D	C	R	M2	105	0	1
2	D	C	R	M2	131	1	0
2	D	C	R	M2	132	1	0
2	D	C	R	M2	…		
2	D	C	R	M2	182	1	0
2	D	C	R	M2	183	1	1
2	D	C	R	M2	184	1	1
2	D	C	R	M2	…		
2	D	C	R	M2	235	1	1
2	D	C	R	M3	131	0	0
2	D	C	R	M3	132	0	0
2	D	C	R	M3	…		
2	D	C	R	M3	182	0	0
2	D	C	R	M3	183	0	1
2	D	C	R	M3	184	0	1
2	D	C	R	M3	…		
2	D	C	R	M3	235	0	1

Figure W2. Visual Representation of Data Structure



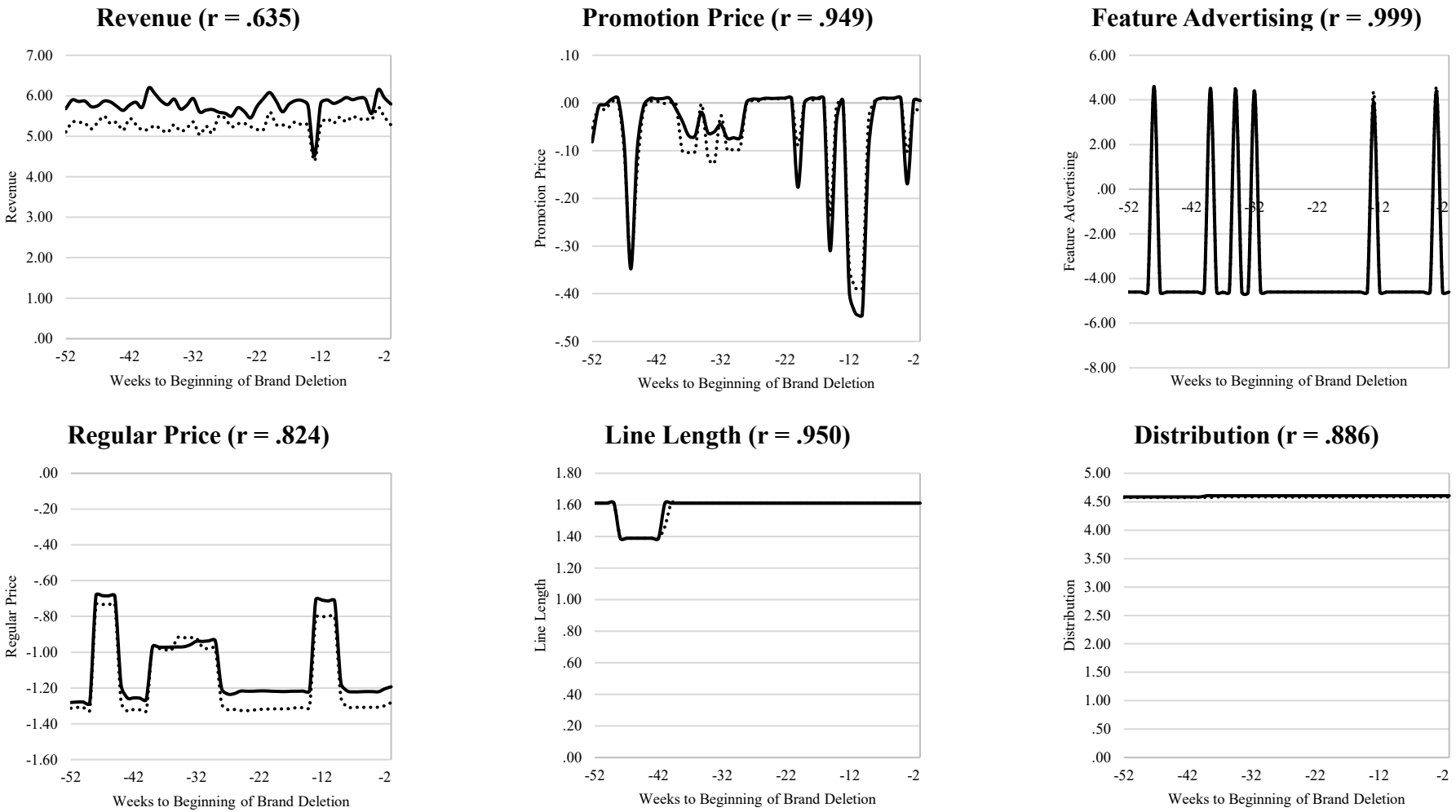
Web Appendix E: Evolution of Treated and Control Units

Figure W3. Pretreatment Evolution for Sample Case at 75th Percentile in Terms of Average Correlation (r = .903)



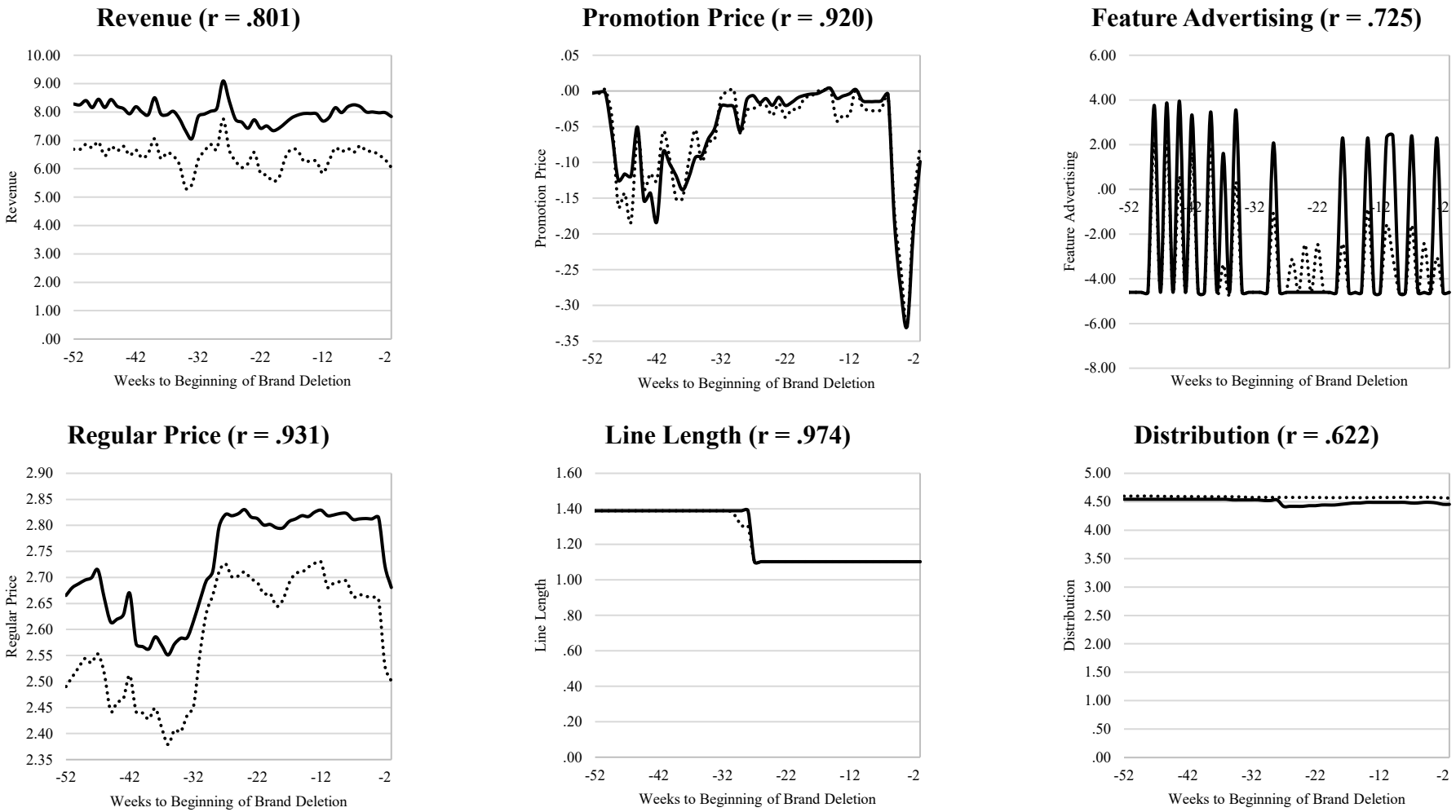
Notes: Solid black lines show the treated unit, dotted black line the average control units.

Figure W4. Pretreatment Evolution for Sample Case at Median in Terms of Average Correlation (r = .868)



Notes: Solid black lines show the treated unit, dotted black line the average control units.

Figure W5. Pretreatment Evolution for Sample Case at 25th Percentile in Terms of Average Correlation (r = .840)



Notes: Solid black lines show the treated unit, dotted black line the average control units.

Web Appendix F: Parameter Estimates of Pre-Trend Deviations Test

Variable	Estimate	SE	p-val.
Weeks to Beginning of Deletion			
2	.005	.002	.014
3	.004	.002	.080
4	.007	.002	.005
5	.004	.003	.115
6	.002	.003	.388
7	.003	.003	.269
8	.002	.003	.522
9	.001	.003	.668
10	-.001	.003	.766
11	.005	.003	.133
12	.003	.003	.468
13	.001	.004	.755
14	.001	.004	.683
15	-.003	.004	.467
16	-.005	.004	.177
17	-.007	.004	.077
18	-.006	.004	.142
19	-.003	.004	.455
20	-.002	.004	.716
21	.001	.004	.794
22	.003	.004	.542
23	.003	.004	.466
24	.003	.005	.468
25	.004	.004	.345
26	.000	.005	.937
27	.002	.005	.682
28	.003	.005	.535
29	.000	.005	.946
30	.002	.005	.628
31	.002	.005	.707
32	.000	.005	.952
33	.003	.005	.473
34	.002	.005	.688
35	.002	.005	.730
36	-.001	.005	.876
37	.004	.005	.453
38	.000	.005	.925
39	-.004	.005	.417
40	-.004	.005	.418
41	-.003	.005	.630
42	.002	.005	.767
43	.003	.005	.516
44	.002	.005	.750
45	.001	.005	.862
46	.005	.005	.325
47	.005	.005	.337
48	.005	.005	.328
49	.006	.005	.288
50	.008	.005	.117
51	.002	.006	.759
52	.001	.006	.847
Cross-section FE	✓		
Time FE	✓		

Notes: Clustered standard errors by panel (market-retailer-category-deleted brand-competing brand combination).
SE = standard error.
N = 23,032,620

Web Appendix G: Parameter Estimates of Control Function Approach

$$EndoVar_{bcmrt} = \underbrace{\alpha_{bcmr} + \alpha_t}_{\text{fixed effects}} + \underbrace{\varphi_0 INST\ TreatPost_{bcmrt}}_{\substack{\text{instrument} \\ \text{brand deletion}}} + \sum_{k=1}^{5} \left\{ \underbrace{\varphi_k INST\ MKTG_{k,bcmrt}}_{\substack{\text{instruments} \\ \text{marketing mix}}} \right\} + \sum_{j=1}^{2} \underbrace{\delta_j Controls}_{\text{competitor price and promotion}} + \varepsilon_{bcmrt} \quad \text{(W1)}$$

$EndoVar_{bcmrt}$ is either $TreatPost_{pcmrt}$ or one of the five marketing variables, $MKTG_{k,bcmrt}$.

Dependent Variable	TreatPost			Promotion Price			Feature Advertising			Regular Price			Line Length			Distribution		
Covariate	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.
TreatPost (IV)	**.513**	**.002**	**.000**	.000	.000	.724	.032	.003	.000	.006	.000	.000	.000	.001	.920	.001	.000	.062
Promotion Price (IV)	-.002	.001	.199	**.914**	**.010**	**.000**	-1.163	.023	.000	-.035	.006	.000	-.031	.005	.000	-.004	.001	.000
Feature Advertising (IV)	.000	.000	.163	-.001	.000	.000	**.722**	**.002**	**.000**	.000	.000	.000	.000	.000	.000	.000	.000	.000
Regular Price (IV)	-.005	.001	.000	-.007	.001	.000	-.033	.006	.000	**.885**	**.004**	**.000**	.006	.002	.001	.000	.001	.837
Line Length (IV)	-.004	.001	.000	-.002	.001	.000	.028	.004	.000	.009	.001	.000	**.942**	**.002**	**.000**	.013	.002	.000
Distribution (IV)	-.007	.002	.001	-.003	.001	.000	-.005	.009	.588	.018	.004	.000	-.006	.006	.274	**.774**	**.019**	**.000**
Competitor Marketing Mix: Promotion Price	-.006	.002	.000	.038	.001	.000	.137	.019	.000	-.002	.001	.175	-.003	.002	.047	.020	.001	.000
Competitor Marketing Mix: Regular Price	-.009	.001	.000	.000	.000	.019	-.008	.004	.063	.013	.002	.000	.007	.002	.000	.001	.001	.195
Cross-section FE	✓			✓			✓			✓			✓			✓		
Time FE	✓			✓			✓			✓			✓			✓		
Sanderson-Windmeijer Test of Instrument Strength F(6, 442934) =	9,945.57		.000	39,652.82		.000	42,565.78		.000	9,162.44		.000	63,609.60		.000	351.52		.000
Correlation of instrument with endogenous variable	.5767		.000	.8737		.000	.7519		.000	.8337		.000	.8656		.000	.6437		.000

Notes: Clustered standard errors by category-retailer-deleted brand-competing brand combination.
SE = Standard error.
N = 55,876,309.
We use boldface for the parameter estimates for the focal instrumental variable for each endogenous variable, highlighting that each of them has a strong positive effect on the corresponding endogenous variable.

Web Appendix H: Category Revenue Model

To assess the category revenue implications of a brand deletion, we aggregate the data from all 14,462 market-retailer-category-deleted brand combinations across competing brands to the market-retailer-category-week level. Irrespective of whether brands meet the sample criteria in our focal analysis, revenue and line length are the sum across all competing brands, category price is value sales divided by volume sales, and feature advertising is the average share of SKUs (by brand) that is being featured in a given week.

We perform the same aggregation for all potential control units (market-retailer-category combinations where the deleted brand was still active in) and calculate the average pre-deletion correlation between each treated and potential control case for the four outcome variables (revenue, category price, category feature advertising, and category line length). We retain the ten control cases with the highest pre-deletion correlation as long as it is > .8, as in the focal analysis. This leads to 6,144 treated cases at the category-market-retailer level with 3.5 control cases on average (N = 3,490,503); thus, the set of treated and control cases does not fully overlap with the analyses at the brand level.

We estimate a similar control function for the endogeneity of the TreatPost term with the same instrument. The instrument is positively and significantly related to the TreatPost variable ($b = .573$, $SE = .007$, $p < .001$).

Web Appendix I: Parameter Estimates New Brand Entry

We estimate a difference-in-difference model to test whether the number of new brands (per annum) changes when a brand is deleted in category c, market m, retailer r, comparing period t=1 for period before (52 weeks) and t=2 for period after the start of the brand deletion (deletion week +52 weeks):

$$NewBrandCount_{cmrt} = \underbrace{\alpha_{cmr}}_{\text{fixed effects}} + \underbrace{\beta_1 TreatPost_{cmrt}}_{\text{change due to brand deletion}} + \underbrace{\delta\, Post_{cmrt}}_{\text{pre.vs.post changes}} + \varepsilon_{cmrt} \quad \text{(W2)}$$

We use three different operationalizations of $NewBrandCount_{cmrt}$: the per-annum number of new brands (i) to the category (ii) to the market and (iii) to the market-retailer-category combination. The results are shown in the table below, indicating no significant effects of brand deletion on the number of new brands across the three operationalizations, with p-values of .564, .205, and .214, respectively,

Table W3. Parameter Estimates New Brand Entry

	DV = New Brands to Category Per Annum (i)			DV = New Brands to Market Per Annum (ii)			DV = New Brands to Market-Retailer Pair Per Annum (iii)		
Parameter	Est.	SE	p-val.	Est.	SE	p-val.	Est.	SE	p-val.
TreatPost	-.013	.023	.564	.044	.034	.205	.049	.039	.214
Post	-.097	.023	.000	-.179	.035	.000	-.242	.040	.000
Constant	.141	.011	.000	1.339	.017	.000	2.019	.019	.000
Cross-Section FE	✓			✓			✓		

Notes: Clustered standard errors by retailer-category combination. DV = dependent variable. SE = standard error. N = 313,650.

Web Appendix J: Parameter Estimates Gradual Model Build Up

	Treatment Effect Only by Brand Type (1)			Add Marketing Mix (2)			Add Control Function Terms (3)			Add Moderators (Focal Model) (4)		
Variables	**Est.**	**SE**	**p-val.**	**Est.**	**SE**	**p-val.**	**Est.**	**SE**	**p-val.**	**Est.**	**SE**	**p-val.**
Intercept shifts												
Treat x Post x …												
Rival National Brand	-.013	.003	.000	-.003	.003	.312	.008	.008	.303	.004	.009	.664
Sister Brand	.000	.011	.997	-.009	.010	.357	.002	.012	.870	.009	.016	.588
Private Label	.000	.007	.962	.008	.006	.156	.020	.009	.028	.021	.010	.035
Post x …												
Rival National Brand	-.009	.005	.096	-.003	.005	.527	-.006	.005	.211	-.012	.006	.034
Sister Brand	-.043	.016	.009	-.026	.013	.038	-.031	.013	.014	-.016	.015	.271
Private Label	-.018	.011	.092	-.030	.009	.001	-.030	.009	.001	-.030	.010	.004
Moderators												
Treat x Post x …												
Sister Brand Present x Rival National Brand										-.004	.006	.495
Sister Brand Present x Private Label										.017	.012	.171
Private Label Present x Rival National Brand										.003	.007	.638
Private Label Present x Sister Brand										-.010	.021	.631
Price Similarity										.028	.011	.010
Deleted Brand Share										.246	.074	.001
Competing Brand Share										.034	.021	.106
Category Concentration										-.158	.031	.000
Category Growth										-.018	.456	.968
Post x …												
Sister Brand Present x Rival National Brand										.007	.007	.329
Sister Brand Present x Private Label										-.006	.016	.691
Private Label Present x Rival National Brand										.011	.006	.080
Private Label Present x Sister Brand										-.026	.015	.087
Price Similarity										-.045	.014	.001
Deleted Brand Share										-.533	.084	.000
Competing Brand Share										-.049	.031	.108
Category Concentration										.128	.048	.008
Category Growth										2.637	.473	.000
Marketing Mix												
Rival National Brand x …												
Promotion Price				-1.804	.018	.000	-1.765	.019	.000	-1.765	.019	.000
Feature Advertising				.026	.000	.000	.034	.000	.000	.034	.000	.000
Regular Price				.113	.027	.000	-.049	.025	.048	-.049	.024	.046
Line Length				1.199	.022	.000	1.263	.025	.000	1.266	.025	.000
Distribution				2.318	.051	.000	2.946	.065	.000	2.942	.065	.000
Sister Brand x …												
Promotion Price				-1.725	.038	.000	-1.684	.038	.000	-1.685	.038	.000
Feature Advertising				.030	.001	.000	.037	.001	.000	.037	.001	.000
Regular Price				-.097	.108	.370	-.259	.104	.013	-.256	.104	.014
Line Length				1.266	.109	.000	1.324	.107	.000	1.325	.107	.000
Distribution				2.456	.207	.000	3.100	.163	.000	3.098	.163	.000
Private Label x …												

Promotion Price		-1.657	.080	.000	-1.589	.080	.000	-1.591	.080	.000
Feature Advertising		.013	.001	.000	.021	.001	.000	.021	.001	.000
Regular Price		-.043	.023	.064	-.176	.026	.000	-.179	.026	.000
Line Length		1.130	.044	.000	1.185	.044	.000	1.190	.044	.000
Distribution		2.908	.231	.000	3.593	.228	.000	3.586	.227	.000
Competitor Marketing Mix										
Promotion Price		.067	.015	.000	.045	.016	.004	.054	.016	.001
Regular Price		.038	.016	.015	.050	.016	.001	.049	.016	.002
Control Function Terms										
TreatPost					-.013	.010	.198	-.009	.010	.331
Promotion Price					.035	.032	.265	.035	.032	.272
Feature Advertising					-.018	.000	.000	-.018	.000	.000
Regular Price					.506	.043	.000	.506	.043	.000
Line Length					-.336	.026	.000	-.337	.026	.000
Distribution					-1.060	.071	.000	-1.057	.071	.000
Cross-section FE	✓	✓			✓			✓		
Time FE	✓	✓			✓			✓		
Likelihood-ratio test compared to (1)		χ^2 (17) = 12,498,355.09		.000	χ^2 (23) = 12,969,537.01		.000	χ^2 (41) = 13,000,569.96		.000
Likelihood-ratio test compared to model to the left		χ^2 (17) = 12,498,355.09		.000	χ^2 (6) = 471,181.92		.000	χ^2 (18) = 31,032.95		.000

Notes: Clustered standard errors by retailer-category-deleted brand-competing brand combination.
SE = standard error.
N = 55,876,309.

Web Appendix K: Parameter Estimates Fraction of New SKUs

Dependent Variable = Brand Revenue	Focal Model			Fraction of New-to-Category SKUs		
Variables	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**
Intercept shifts						
Treat x Post x …						
Rival National Brand	.004	.009	.664	.004	.009	.640
Sister Brand	.009	.016	.588	.009	.016	.585
Private Label	.021	.010	.035	.020	.010	.048
Post x …						
Rival National Brand	-.012	.006	.034	-.014	.006	.019
Sister Brand	-.016	.015	.271	-.017	.015	.241
Private Label	-.030	.010	.004	-.024	.010	.024
Moderators						
Treat x Post x …						
Sister Brand Present x Rival National Brand	-.004	.006	.495	-.004	.006	.529
Sister Brand Present x Private Label	.017	.012	.171	.017	.012	.177
Private Label Present x Rival National Brand	.003	.007	.638	.003	.007	.682
Private Label Present x Sister Brand	-.010	.021	.631	-.010	.021	.651
Price Similarity	.028	.011	.010	.028	.011	.009
Deleted Brand Share	.246	.074	.001	.231	.074	.002
Competing Brand Share	.034	.021	.106	.034	.021	.104
Category Concentration	-.158	.031	.000	-.156	.031	.000
Category Growth	-.018	.456	.968	-.013	.457	.977
Fraction of New SKUs				.043	.050	.386
Post x …						
Sister Brand Present x Rival National Brand	.007	.007	.329	-.007	.016	.646
Sister Brand Present x Private Label	-.006	.016	.691	.006	.007	.417
Private Label Present x Rival National Brand	.011	.006	.080	.011	.006	.068
Private Label Present x Sister Brand	-.026	.015	.087	-.028	.015	.069
Price Similarity	-.045	.014	.001	-.046	.014	.001
Deleted Brand Share	-.533	.084	.000	-.050	.031	.106
Competing Brand Share	-.049	.031	.108	-.515	.085	.000
Category Concentration	.128	.048	.008	.124	.048	.009
Category Growth	2.637	.473	.000	2.630	.473	.000
Fraction of New SKUs				-.245	.059	.000
Marketing Mix						
Rival National Brand x …						
Promotion Price	-1.765	.019	.000	-1.765	.019	.000
Feature Advertising	.034	.000	.000	.034	.000	.000
Regular Price	-.049	.024	.046	-.047	.024	.053
Line Length	1.266	.025	.000	1.268	.025	.000
Distribution	2.942	.065	.000	2.941	.065	.000
Sister Brand x …						
Promotion Price	-1.685	.038	.000	-1.685	.038	.000
Feature Advertising	.037	.001	.000	.037	.001	.000
Regular Price	-.256	.104	.014	-.256	.104	.014

Line Length	1.325	.107	.000	1.327	.107	.000
Distribution	3.098	.163	.000	3.097	.163	.000
Private Label x …						
Promotion Price	-1.591	.080	.000	-1.589	.080	.000
Feature Advertising	.021	.001	.000	.021	.001	.000
Regular Price	-.179	.026	.000	-.171	.026	.000
Line Length	1.190	.044	.000	1.198	.044	.000
Distribution	3.586	.227	.000	3.582	.227	.000
Competitor Marketing Mix						
Promotion Price	.054	.016	.001	.056	.016	.000
Regular Price	.049	.016	.002	.049	.016	.002
Control Function Terms						
Treat x Post	-.009	.010	.331	-.009	.010	.336
Promotion Price	.035	.032	.272	.035	.032	.270
Feature Advertising	-.018	.000	.000	-.018	.000	.000
Regular Price	.506	.043	.000	.505	.043	.000
Line Length	-.337	.026	.000	-.337	.026	.000
Distribution	-1.057	.071	.000	-1.056	.071	.000
Cross-section FE	✓			✓		
Time FE	✓			✓		
Correlation between estimates and corresponding focal model estimates	n/a			>.999		

Notes: Clustered standard errors by retailer-category-deleted brand-competing brand combination.
SE = standard error, p = p-value.
N = 55,876,309

Web Appendix L: National Distribution Elasticities

We convert the retailer-specific distribution elasticity to national distribution elasticities by assessing the change in national revenue and the change in national distribution in response to a 1% change in distribution at the focal retailer. Specifically, we calculate:

$$NatDistrElasticity = \frac{\Delta NatRevenue\ in\ \%}{\Delta NatDistribution\ in\ \%} \quad \text{(W3)}$$

which is the percentage change in nation-wide brand revenue divided by the percentage change in national distribution.

We can calculate the numerator of (W3) as

$$\Delta NatRevenue\ in\ \% = \frac{RetRevenue^{After} - RetRevenue^{Before}}{RetRevenue^{Before} + OtherRetRevenue}$$

where $RetRevenue^{After}$ is brand revenue at the focal retailer after the 1% lift in distribution at that retailer, which can be calculated as brand revenue at that retailer before the lift ($RetRevenue^{Before}$) times the multiplier resulting from the 1% lift in distribution using the estimated retailer distribution elasticity ($RetDistrElasticity$):

$$RetRevenue^{After} = RetRevenue^{Before} \times (1 + 1\%\ x\ RetDistrElasticity)$$

and where $OtherRetRevenue$ is aggregate brand revenue at other retailers nationally.

We can calculate the denominator of (W3) as

$$\Delta NatDistribution\ in\ \% = \frac{NatDistribution^{After} - NatDistribution^{Before}}{NatDistribution^{Before}}$$

where $NatDistribution^{Before}$ is the national distribution before the 1% lift in distribution at the focal retailer, which equals the weighted average of the focal retailer's distribution of the focal brand before the 1% distribution lift ($RetDistribution^{Before}$) and the (unchanged) brand distribution at other retailers nationally ($OtherRetDistribution$), using the focal retailer's market share of the brand's national revenue $RetMarketShare$ (and 1 minus this share) as weights:

$$NatDistribution^{Before} = RetMarketShare\ x\ RetDistribution^{Before} + (1 - RetMarketShare)\ x\ OtherRetDistribution$$

Similarly, $NatDistribution^{After}$ is the national distribution after the 1% lift in distribution at the focal retailer, which equals the weighted average of the focal retailer's distribution of the focal brand after the 1% distribution lift ($RetDistribution^{After} = RetDistribution^{Before} \times (1 + 1\%)$) and the (unchanged) distribution at other retailers ($OtherRetDistribution$), again using the focal retailer's market share of the brand's national revenue $RetMarketShare$ (and 1 minus this share) as weights:

$$NatDistribution^{After} = RetMarketShare\ x\ RetDistribution^{After} + (1 - RetMarketShare)\ x\ OtherRetDistribution$$

Using Equation W3 and average values for the other variables (e.g., retailer market shares) from our sample, we convert the estimated retailer-level distribution elasticities to national distribution elasticities. Table W4 presents the elasticities:

Table W4. Distribution Elasticity Conversion Results

Brand Type	$RetDistrElasticity$	$NatDistrElasticity$
Rival National Brand	2.942	1.103
Sister Brand	3.098	2.015
Private Label	3.586	.316

Web Appendix M: Net Revenue Effect for Manufacturers of Deleted Brands

	Brand	Brand's Revenue Before	Recouped Revenue	Share Recouped Revenue	Avg. # of Sister Brands
1	Pediasure	$ 22,312,361	$ 2,237,951	10.03%	1.41
2	Wisk	$ 7,134,366	$ 2,035,075	28.52%	3.79
3	Maalox	$ 4,442,481	$ 151,947	3.42%	1.00
4	Altoids	$ 3,376,014	$ 448,786	13.29%	1.75
5	Mousse	$ 2,533,656	$ 677,681	26.75%	1.08
6	Feline	$ 2,488,042	$ 591,744	23.78%	1.00
7	Planters	$ 2,452,799	$ 605,442	24.68%	1.02
8	Eureka!	$ 2,148,220	$ 999,052	46.51%	1.38
9	Mission Life Balance Plus!	$ 1,711,848	$ 3,467,972	202.59%	2.38
10	Waggin'	$ 1,479,048	$ 121,210	8.20%	1.00
…					
25	Earth	$ 848,996	$ 5,384	.63%	1.00
26	Whisker	$ 794,058	$ 947,404	119.31%	1.02
27	Honey Graham Oh!s	$ 770,531	$ 101,148	13.13%	1.14
28	Wonderful Almond Accents	$ 766,378	$ 221,213	28.86%	1.00
29	Irish	$ 744,436	$ 611,397	82.13%	1.09
30	Rana	$ 742,649	$ 165,954	22.35%	1.00
…					
50	Marie	$ 477,413	$ 52,784	11.06%	1.23
…					
100	Esskay	$ 160,523	$ 18,718	11.66%	1.00
…					
215	Mentadent	$ 5,258	$ 14	.26%	1.00
Average		**$ 485,799**	**$ 183,453**	**37.76%**	**1.36**

Notes: Revenues are averages per year across market-retailer pairs where at least one sister brand is active. As such these values differ from those in Table 2 in the main text.

Web Appendix N: Robustness Checks

We conduct several robustness checks to establish the sensitivity of our findings to modeling decisions, which we describe in more detail below. We report the results in Table WA5 of this Web Appendix.

First, we empirically determine the beginning of a brand's deletion using changes in the (moving-average) distribution as noted in the "Data and Measures" section. As we are using a moving average, the brand deletion may have started earlier than detected by our decision rule. To assess this possibility, we set T_0 (the beginning of the deletion) to two weeks before the week in our focal analyses and re-estimate our models.

Second, we leverage a staggered adoption setting with a stacked DID approach. In the presence of heterogeneous effects, a standard two-way fixed effects (TWFE) regression coefficient may lead to biased treatment effect estimates (Baker, Larcker, and Wang 2022; Goodman-Bacon 2021). In that case, the standard TWFE regression does not offer a single clean DID estimate, but a weighted average of three types of comparisons: (i) treated versus never-treated, (ii) early-treated versus later-treated (while later is not yet treated), and (iii) later-treated versus earlier-treated (while earlier is already treated), arising because TWFE mechanically uses all timing variation as identifying variation. While contrasts (i) and (ii) are proper identification sources, source (iii) (later-vs-earlier contrasts) is problematic: it compares newly-treated groups to already-treated groups, which are no longer valid controls when treatment effects are heterogeneous. As detailed in the Method section, we address this issue by using as controls only cases that are never treated within the time of the treated case. By construction, this removes later-vs-earlier contrasts, resolving the issue.

As an alternative, we adopt a novel and computationally efficient two-stage DID estimation approach as a robustness check (Gardner et al. 2024). The overarching idea is to first remove underlying group and time patterns using untreated observations and then estimate treatment effects by

comparing treated outcomes to these adjusted baselines, thereby recovering an interpretable average effect even when adoption is staggered and effects vary across groups and periods. Because earlier-treated units (part of later-vs-earlier contrasts) never inform the baselines, source (iii) (later-vs-earlier) contrasts disappear. Specifically, we first regress the outcome variable (revenue) on cross-section and time fixed effects, the marketing mix, and moderators based on untreated data. We then subtract these fixed effects from the outcome variable to obtain an adjusted outcome. In a second step, we regress the adjusted outcome on a treatment indicator and its interaction effects with the moderator variables. While this procedure does not deliver marketing mix effectiveness estimates based on the full sample, it isolates the treatment effect in a way that remains valid in the presence of treatment effect heterogeneity.

Third, we use controls to create a counterfactual for each brand's evolution. While we imposed a minimum average correlation of $\bar{\rho} > .80$ between revenue and marketing mix trajectories for the treated and control brands (see Table 4), we also investigated whether imposing higher minimum average correlations of $\bar{\rho} > .85$ or $\bar{\rho} > .90$ makes a difference. Using these higher thresholds reduces the number of treatment-control pairs because there are fewer pairs meeting these thresholds, from N = 55,876,309 for our default minimum $\bar{\rho} > .80$, to N = 43,855,533 for $\bar{\rho} > .85$ and N = 28,157,212 for $\bar{\rho} > .90$.

Fourth, while our focal model uses clustered errors at the retailer-category-deleted brand-competing brand level, we also implement clustered standard errors at the panel (i.e., market-retailer-category-deleted brand-competing brand) level as an alternative to account for autocorrelation only.

Fifth, we assess to what extent our results in the revenue model are driven by the corrections for endogeneity. To this end, we estimate three additional versions of Equation 1: without any control-function terms, with only the control-function term for the deletion decision, and with only control-

function terms for the marketing mix.

Sixth, a brand deletion is typically staggered across markets and retailers over months or even years. Our difference-in-differences approach capitalizes on the spread-out nature of brand deletions by using market-retailer-category combinations where the brand has not been deleted yet as controls for cases where the brand deletion has already taken place. Consequently, cases where the brand deletion occurs last lack valid controls and are excluded from the analysis. However, a relevant question is whether effects differ between early versus late brand deletion cases. To investigate, we divide the sample into "Early deletion" cases (start date before the median) and "Late deletion" cases (after the median). We note that the average number of controls for the "Early deletion" group (6.12) closely matches that for the "Late deletion" group (5.96), with no significant differences. Similarly, the average pretreatment-control correlation for the Early deletion group (.896) does not materially differ (p=.144) from the Late deletion group (.893). We also add an interaction effect of treatment x post with an indicator for Early deletion (base = Late deletion) in Equation 1. We find that the added term does not have a significant effect (b=-.000, p=.949), suggesting that the Early deletion group does not benefit differentially from the brand deletion than the Late deletion group. The focal parameter estimates are not affected.

Seventh, it is conceivable that the five marketing mix variables have interactive effects and that there are decreasing returns to scale for line length. We added all 10 pairwise interactions to Equation 1, and separately added squared log line length, as reported in Table WA5. While several of these additional terms are significant, the focal estimates preserve their sign and significance. For parsimony reasons, we stick to the focal model.

Table WA5: Detailed Parameter Estimates Focal Model and Robustness Checks

Dependent Variable = Brand Revenue	Focal Model			Pre-treatment Corr. > .85			Pre-treatment Corr. > .90			Standard Errors Clustered at Panel			Gardner approach			Brand Deletion 2 Weeks Earlier			Early vs. Late Deletion			Marketing Mix Interaction Terms			Squared Line Length			No Control Function			No Control Function Marketing Mix			No Control Function Treat x Post		
Variables	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**	**Est.**	**SE**	**p**
Intercept shifts																																			
Treat x Post x …																																			
Rival National Brand	.004	.009	.664	.005	.011	.601	.000	.014	.976	.004	.009	.659	.006	.007	.421	.005	.009	.567	.004	.010	.695	.004	.009	.669	-.001	.009	.873	-.005	.006	.370	-.001	.009	.934	-.003	.006	.619
Sister Brand	.009	.016	.588	.017	.018	.344	.029	.022	.185	.009	.020	.661	.009	.021	.666	.013	.016	.393	.008	.016	.628	.009	.016	.586	.004	.016	.819	-.006	.015	.672	-.002	.016	.925	.002	.015	.910
Private Label	.021	.010	.035	.027	.011	.017	.014	.015	.350	.021	.013	.088	.034	.010	.001	.030	.010	.002	.021	.010	.044	.021	.010	.034	.017	.010	.098	.011	.007	.127	.015	.010	.135	.014	.007	.042
Post x …																																			
Rival National Brand	-.012	.006	.034	-.013	.007	.059	-.011	.009	.231	-.012	.003	.000				-.005	.005	.380	-.008	.007	.258	-.012	.006	.032	-.011	.006	.055	-.012	.006	.039	-.012	.006	.038	-.012	.006	.035
Sister Brand	-.016	.015	.271	-.020	.017	.246	-.005	.023	.811	-.016	.007	.023				.002	.014	.878	-.011	.015	.452	-.017	.015	.267	-.015	.015	.319	-.017	.015	.263	-.017	.015	.255	-.016	.015	.284
Private Label	-.030	.010	.004	-.026	.011	.018	-.020	.013	.121	-.030	.004	.000				-.023	.010	.025	-.025	.011	.020	-.030	.010	.003	-.028	.010	.007	-.028	.010	.006	-.028	.010	.006	-.030	.010	.004
Moderators																																			
Treat x Post x …																																			
Sister Brand Present x Rival National Brand	-.004	.006	.495	-.007	.006	.303	-.006	.008	.465	-.004	.009	.632	.000	.009	.962	-.002	.006	.724	-.004	.006	.509	-.005	.006	.461	-.005	.006	.423	-.003	.006	.575	-.003	.006	.583	-.004	.006	.484
Sister Brand Present x Private Label	.017	.012	.171	-.001	.013	.922	.013	.019	.498	.017	.017	.323	.003	.017	.874	.003	.012	.800	.017	.012	.161	.016	.012	.178	.013	.012	.275	.018	.012	.144	.018	.012	.141	.017	.012	.175
Private Label Present x Rival National Brand	.003	.007	.638	.004	.008	.624	.019	.010	.067	.003	.009	.702	.007	.009	.465	.003	.007	.656	.004	.007	.630	.004	.007	.604	.003	.007	.687	.004	.007	.607	.004	.007	.605	.003	.007	.640
Private Label Present x Sister Brand	-.010	.021	.631	-.011	.023	.620	-.002	.025	.945	-.010	.027	.709	-.008	.027	.767	-.015	.020	.467	-.009	.021	.667	-.010	.021	.633	-.013	.021	.539	-.001	.021	.945	-.002	.021	.942	-.010	.021	.634
Price Similarity	.028	.011	.010	.022	.012	.065	.026	.016	.099	.028	.016	.088	.029	.016	.069	.024	.010	.019	.028	.011	.010	.028	.011	.010	.027	.011	.013	.031	.011	.005	.031	.011	.004	.027	.011	.011
Deleted Brand Share	.246	.074	.001	.296	.089	.001	.443	.119	.000	.246	.092	.007	.283	.086	.001	.252	.073	.001	.242	.074	.001	.248	.074	.001	.247	.073	.001	.266	.074	.000	.266	.074	.000	.247	.074	.001
Competing Brand Share	.034	.021	.106	.052	.024	.028	.146	.033	.000	.034	.032	.295	-.001	.033	.973	.019	.019	.332	.034	.021	.105	.033	.021	.117	.033	.021	.118	.032	.021	.130	.032	.021	.130	.034	.021	.105
Category Concentration	-.158	.031	.000	-.198	.036	.000	-.286	.052	.000	-.158	.048	.001	-.108	.050	.032	-.123	.029	.000	-.158	.031	.000	-.157	.031	.000	-.158	.031	.000	-.147	.032	.000	-.147	.032	.000	-.159	.031	.000
Category Growth	-.018	.456	.968	.268	.468	.567	.058	.497	.906	-.018	.655	.978	-.115	.683	.866	-.456	.315	.147	-.027	.456	.953	-.027	.456	.953	.027	.452	.952	-.002	.461	.996	-.005	.461	.991	-.014	.456	.976
Early Market																			.000	.005	.949															
Post x …																																			
Sister Brand Present x Rival National Brand	.007	.007	.329	.009	.008	.252	.008	.011	.454	.007	.003	.024				.009	.007	.179	.006	.007	.376	.007	.007	.317	.011	.007	.124	.013	.007	.057	.013	.007	.058	.007	.007	.320
Sister Brand Present x Private Label	-.006	.016	.691	-.004	.017	.830	-.016	.021	.461	-.006	.007	.350				.004	.016	.797	-.007	.016	.665	-.005	.016	.771	-.002	.015	.875	-.007	.016	.678	-.007	.016	.673	-.006	.016	.698
Private Label Present x Rival National Brand	.011	.006	.080	.008	.007	.246	.001	.010	.901	.011	.003	.001				.012	.006	.050	.011	.006	.070	.011	.006	.081	.012	.006	.051	.012	.006	.041	.012	.006	.042	.011	.006	.079
Private Label Present x Sister Brand	-.026	.015	.087	-.042	.018	.019	-.095	.024	.000	-.026	.009	.005				-.032	.015	.026	-.027	.015	.082	-.027	.015	.083	-.021	.015	.169	-.018	.016	.253	-.018	.016	.256	-.026	.015	.086
Price Similarity	-.045	.014	.001	-.058	.016	.000	-.079	.019	.000	-.045	.006	.000				-.042	.014	.002	-.045	.014	.001	-.046	.014	.001	-.044	.014	.002	-.047	.014	.001	-.047	.014	.001	-.045	.014	.001
Deleted Brand Share	-.533	.084	.000	-.623	.106	.000	-.849	.144	.000	-.533	.038	.000				-.554	.082	.000	-.531	.084	.000	-.532	.084	.000	-.548	.084	.000	-.512	.084	.000	-.512	.084	.000	-.532	.084	.000
Competing Brand Share	-.049	.031	.108	-.102	.036	.004	-.228	.048	.000	-.049	.012	.000				-.022	.026	.401	-.050	.031	.106	-.049	.031	.111	-.040	.031	.190	-.047	.031	.130	-.047	.031	.130	-.049	.031	.108
Category Concentration	.128	.048	.008	.229	.056	.000	.424	.076	.000	.128	.018	.000				.027	.040	.489	.127	.048	.008	.128	.048	.008	.116	.048	.015	.104	.048	.030	.104	.048	.030	.128	.048	.007
Category Growth	2.637	.473	.000	2.287	.508	.000	1.986	.598	.001	2.637	.205	.000				2.567	.319	.000	2.629	.471	.000	2.633	.472	.000	2.652	.474	.000	2.927	.492	.000	2.924	.492	.000	2.642	.473	.000
Early Market																			-.014	.007	.040															
Marketing Mix																																			
Rival National Brand x …																																			
Promotion Price	-1.765	.019	.000	-1.761	.020	.000	-1.759	.023	.000	-1.765	.008	.000				-1.769	.018	.000	-1.765	.019	.000	-1.791	.022	.000	-1.768	.019	.000	-1.805	.018	.000	-1.805	.018	.000	-1.765	.019	.000
Feature Advertising	.034	.000	.000	.034	.000	.000	.035	.001	.000	.034	.000	.000				.035	.000	.000	.034	.000	.000	.034	.000	.000	.034	.000	.000	.026	.000	.000	.026	.000	.000	.034	.000	.000
Regular Price	-.049	.024	.046	-.054	.029	.061	-.086	.035	.013	-.049	.008	.000				.044	.028	.118	-.049	.024	.047	-.034	.025	.170	-.045	.024	.063	.113	.027	.000	.113	.027	.000	-.049	.024	.046
Line Length	1.266	.025	.000	1.303	.029	.000	1.392	.038	.000	1.266	.009	.000				1.253	.025	.000	1.266	.025	.000	1.266	.025	.000	1.195	.023	.000	1.202	.022	.000	1.202	.022	.000	1.266	.025	.000
Distribution	2.942	.065	.000	2.977	.079	.000	3.040	.109	.000	2.942	.039	.000				3.009	.066	.000	2.942	.065	.000	2.988	.073	.000	2.792	.065	.000	2.315	.051	.000	2.315	.051	.000	2.942	.065	.000
Sister Brand x …																																			
Promotion Price	-1.685	.038	.000	-1.705	.041	.000	-1.734	.048	.000	-1.685	.015	.000				-1.677	.038	.000	-1.685	.038	.000	-1.710	.038	.000	-1.687	.038	.000	-1.725	.038	.000	-1.725	.038	.000	-1.685	.038	.000
Feature Advertising	.037	.001	.000	.038	.001	.000	.038	.001	.000	.037	.000	.000				.038	.001	.000	.037	.001	.000	.037	.001	.000	.037	.001	.000	.030	.001	.000	.030	.001	.000	.037	.001	.000
Regular Price	-.256	.104	.014	-.261	.116	.025	-.300	.145	.038	-.256	.036	.000				-.113	.104	.279	-.256	.104	.014	-.254	.104	.014	-.240	.105	.022	-.094	.108	.384	-.094	.108	.384	-.256	.104	.014
Line Length	1.325	.107	.000	1.376	.121	.000	1.459	.149	.000	1.325	.038	.000				1.251	.101	.000	1.326	.107	.000	1.323	.107	.000	1.239	.107	.000	1.267	.109	.000	1.267	.109	.000	1.325	.107	.000
Distribution	3.098	.163	.000	3.076	.166	.000	3.122	.188	.000	3.098	.138	.000				3.164	.137	.000	3.097	.163	.000	3.153	.175	.000	2.918	.163	.000	2.456	.207	.000	2.456	.207	.000	3.098	.163	.000
Private Label x …																																			
Promotion Price	-1.591	.080	.000	-1.433	.072	.000	-1.360	.070	.000	-1.591	.030	.000				-1.608	.076	.000	-1.591	.080	.000	-1.623	.081	.000	-1.588	.079	.000	-1.659	.080	.000	-1.659	.080	.000	-1.591	.080	.000
Feature Advertising	.021	.001	.000	.022	.001	.000	.023	.001	.000	.021	.000	.000				.023	.001	.000	.021	.001	.000	.021	.001	.000	.021	.001	.000	.013	.001	.000	.013	.001	.000	.021	.001	.000
Regular Price	-.179	.026	.000	-.203	.028	.000	-.213	.032	.000	-.179	.010	.000				-.094	.023	.000	-.179	.026	.000	-.178	.027	.000	-.158	.026	.000	-.046	.023	.044	-.047	.023	.043	-.179	.026	.000
Line Length	1.190	.044	.000	1.159	.047	.000	1.223	.061	.000	1.190	.018	.000				1.181	.044	.000	1.190	.044	.000	1.187	.043	.000	1.135	.044	.000	1.134	.044	.000	1.134	.044	.000	1.189	.044	.000
Distribution	3.586	.227	.000	3.717	.271	.000	3.966	.349	.000	3.586	.150	.000				3.386	.204	.000	3.585	.227	.000	3.659	.221	.000	3.377	.227	.000	2.903	.231	.000	2.903	.231	.000	3.586	.227	.000
Marketing Mix (Addtl.)																																			
Promotion Price x …																																			
Feature Advertising																						.009	.003	.000												
Regular Price																						.788	.179	.000												
Line Length																						1.206	.098	.000												

Distribution																			-.050	.167	.766														
Feature Advertising x …																																			
Regular Price																			.001	.003	.839														
Line Length																			-.002	.002	.318														
Distribution																			.070	.016	.000														
Regular Price x …																																			
Line Length																			.187	.080	.020														
Distribution																			.046	.092	.615														
Line Length x …																																			
Distribution																			.043	.062	.491														
Line Length (Squared)																						-.460	.039	.000											
Competitor Marketing Mix																																			
Promotion Price	.054	.016	.001	.068	.017	.000	.087	.021	.000	.054	.008	.000		.053	.016	.001	.055	.016	.000	.052	.016	.001	.058	.016	.000	.077	.016	.000	.077	.016	.000	.054	.016	.001
Regular Price	.049	.016	.002	.049	.018	.007	.069	.023	.002	.049	.006	.000		.053	.014	.000	.049	.016	.002	.046	.016	.003	.049	.016	.002	.038	.016	.018	.038	.016	.018	.049	.016	.002
Control Function Terms																																			
Treat x Post	-.009	.010	.331	-.006	.011	.600	-.005	.015	.745	-.009	.007	.192		-.011	.009	.224	-.009	.010	.346	-.009	.010	.337	-.001	.010	.889				-.006	.010	.536				
Promotion Price	.035	.032	.272	.010	.036	.789	-.028	.043	.525	.035	.018	.051		.271	.026	.000	.034	.032	.276	.041	.033	.216	.041	.032	.196							.035	.032	.272
Feature Advertising	-.018	.000	.000	-.017	.001	.000	-.017	.001	.000	-.018	.000	.000		-.025	.000	.000	-.018	.000	.000	-.018	.000	.000	-.018	.000	.000							-.018	.000	.000
Regular Price	.506	.043	.000	.571	.051	.000	.652	.065	.000	.506	.016	.000		.229	.025	.000	.506	.043	.000	.504	.042	.000	.496	.043	.000							.506	.043	.000
Line Length	-.337	.026	.000	-.360	.031	.000	-.426	.041	.000	-.337	.015	.000		-.395	.023	.000	-.337	.026	.000	-.335	.026	.000	-.346	.026	.000							-.337	.026	.000
Distribution	-1.057	.071	.000	-1.084	.085	.000	-1.083	.118	.000	-1.057	.059	.000		-1.209	.071	.000	-1.057	.071	.000	-1.074	.072	.000	-.975	.069	.000							-1.057	.071	.000
Cross-section FE	✓			✓			✓			✓				✓			✓			✓			✓			✓			✓			✓		
Time FE	✓			✓			✓			✓				✓			✓			✓			✓			✓			✓			✓		
Correlation between estimates and corresponding focal model estimates	n/a			.997			.991			1.000			.928	.996			>.999			>.999			>.999			.989			.990			>.999		

Notes: Clustered standard errors by retailer-category-deleted brand-competing brand combination (except for the fourth model, where standard errors are clustered at the market-retailer-category-deleted brand-competing brand combination level).
SE = standard error, p = p-value.
N = 55,876,309 (except for Pre-treatment Corr. > .85: 43,855,533, Pre-treatment Corr. > .90: 28,157,212 and “Brand Deletion 2 Weeks Earlier”: 55,572,757)